

Sri Lanka Institute of Information Technology



BSc (Hons) in Information Technology specialized in Data Science

IT3021 - Data Warehousing and Business Intelligence

Assignment 2

IT23578678

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WE.DS.01.02

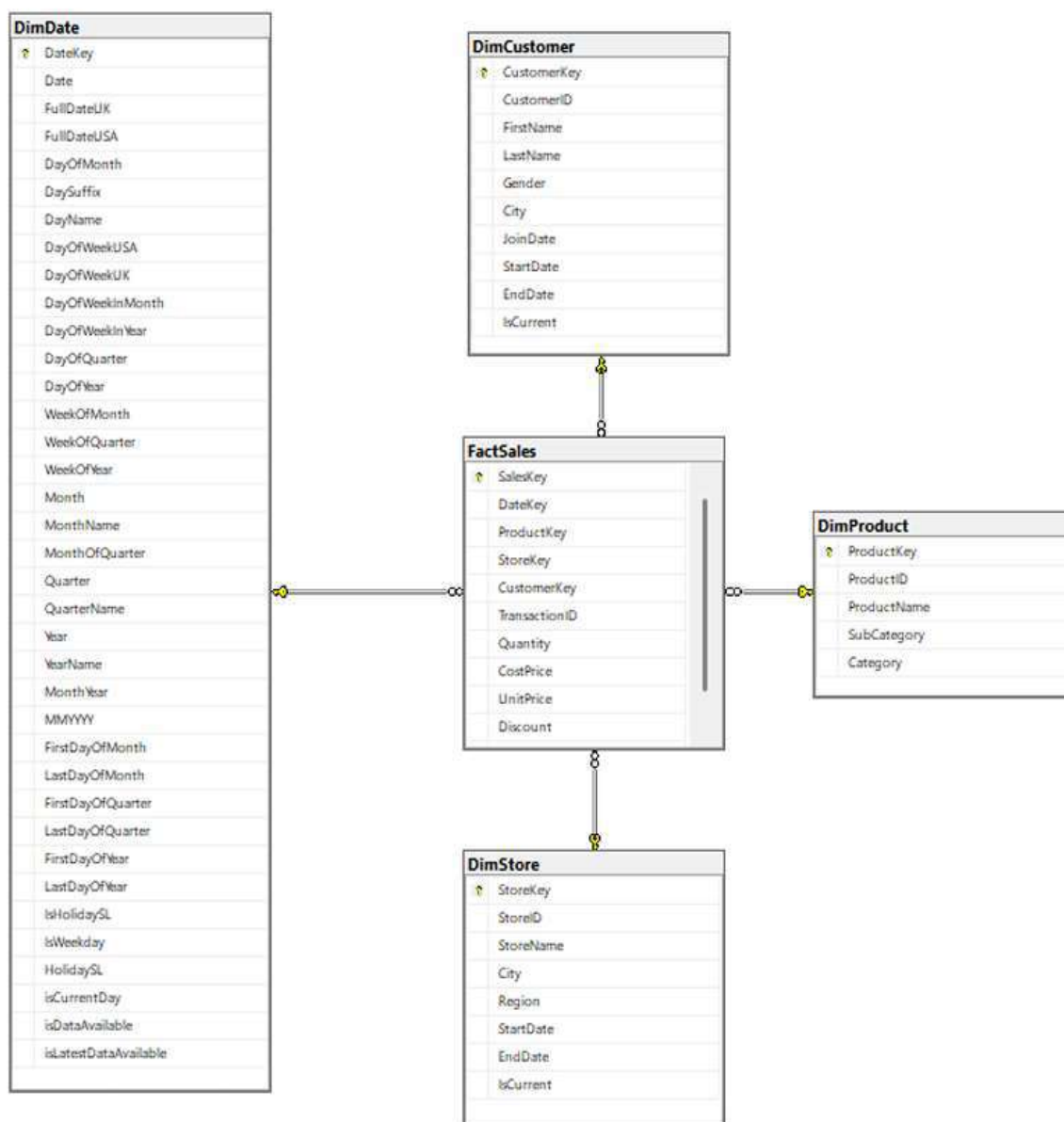
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Step 1: Data source

The data source for this project is the **Retail_DW** data warehouse, which was designed and populated during Assignment 1. The warehouse follows a **Star Schema** architecture. It integrates data from a transactional retail system, covering sales, products, customers, and stores. The primary objective of this data source is to enable multi-dimensional analysis of sales performance across different time periods, regions, and product categories.

Star Schema of Retail_DW



Components of Star Schema

results		Messages													
	SalesKey	DateKey	ProductKey	StoreKey	CustomerKey	TransactionID	Quantity	CostPrice	UnitPrice	Discount	TotalAmount	TotalCost	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hou
1	1	20240705	1	1	51	T01335	1	1323.38	1673.69	0.00	1673.69	1323.38	2026-03-31 21:39:07.080	2026-03-31 15:20:44.000	-6.00
2	2	20240203	1	5	59	T04149	3	1323.38	1673.69	0.00	5021.07	3970.14	2026-03-31 21:39:07.080	2026-04-01 11:12:45.000	14.00
3	3	20241108	1	1	92	T00818	2	1323.38	1673.69	0.15	2845.27	2646.76	2026-03-31 21:39:07.080	2026-03-31 09:15:26.000	-12.00
4	4	20250601	1	2	97	T00824	5	1323.38	1673.69	0.15	7113.18	6616.90	2026-03-31 21:39:07.080	2026-03-31 10:48:22.000	-11.00
5	5	20231217	1	5	188	T01181	2	1323.38	1673.69	0.10	3012.64	2646.76	2026-03-31 21:39:07.080	2026-03-30 15:15:46.000	-30.00
6	6	20240807	1	1	37	T02001	2	1323.38	1673.69	0.00	3347.38	2646.76	2026-03-31 21:39:07.080	2026-03-31 14:14:43.000	-7.00
7	7	20240510	1	2	13	T04949	2	1323.38	1673.69	0.15	2845.27	2646.76	2026-03-31 21:39:07.080	2026-03-30 18:10:56.000	-27.00
8	8	20250808	1	1	151	T03225	4	1323.38	1673.69	0.15	5690.54	5293.52	2026-03-31 21:39:07.080	2026-03-30 19:22:33.000	-26.00
9	9	20240722	1	5	15	T01973	3	1323.38	1673.69	0.00	5021.07	3970.14	2026-03-31 21:39:07.080	2026-03-31 14:25:37.000	-7.00
10	10	20250601	1	5	58	T04119	3	1323.38	1673.69	0.15	4267.90	3970.14	2026-03-31 21:39:07.080	2026-03-30 16:55:33.000	-29.00
11	11	20250417	1	3	90	T04114	4	1323.38	1673.69	0.15	5690.54	5293.52	2026-03-31 21:39:07.080	2026-03-30 18:16:26.000	-27.00
12	12	20240927	1	5	96	T00885	3	1323.38	1673.69	0.15	4267.90	3970.14	2026-03-31 21:39:07.080	2026-03-31 13:45:22.000	-8.00
13	13	20241215	1	2	29	T02752	5	1323.38	1673.69	0.10	7531.60	6616.90	2026-03-31 21:39:07.080	2026-04-01 11:16:45.000	14.00
14	14	20250113	1	2	115	T03890	5	1323.38	1673.69	0.05	7950.02	6616.90	2026-03-31 21:39:07.080	2026-03-30 23:19:29.000	-22.00
15	15	20231003	1	3	101	T04151	2	1323.38	1673.69	0.15	2845.27	2646.76	2026-03-31 21:39:07.080	2026-03-31 18:16:26.000	-3.00
16	16	20240525	1	5	197	T00289	4	1323.38	1673.69	0.00	6694.76	5293.52	2026-03-31 21:39:07.080	2026-03-30 15:32:47.000	-30.00

1. **FactSales**: This is the central table containing quantitative measures such as Quantity, UnitPrice, TotalAmount, TotalCost and the txn_process_time_hours. It contains following foreign keys.

DateKey : References Date in the **DimDate** table

ProductKey : References Product in the **DimProduct** table

CustomerKey : References Customer in the **DimCustomer** table

StoreKey : References Store in the **DimStore** table

	ProductKey	ProductID	ProductName	SubCategory	Category
1	1	P001	Like Camera	Camera	Electronics
2	2	P012	Above Camera	Camera	Electronics
3	3	P007	Understand Camera	Camera	Electronics
4	4	P040	Piece Headphones	Headphones	Fashion
5	5	P011	Either Headphones	Headphones	Fashion
6	6	P017	Find Headphones	Headphones	Fashion
7	7	P019	Traditional Laptop	Laptop	Groceries
8	8	P035	Chair Laptop	Laptop	Groceries
9	9	P034	Everything Laptop	Laptop	Groceries
10	10	P047	Or Smartphone	Smartphone	NA
11	11	P018	Democratic Smartphone	Smartphone	NA
12	12	P024	Him Smartphone	Smartphone	NA
13	13	P028	Deal Smartphone	Smartphone	NA
14	14	P010	Step Smartphone	Smartphone	NA
15	15	P027	Hospital Smartphone	Smartphone	NA
16	16	P002	Audience Television	Television	NA

2. **DimProduct**: Contains descriptive attributes including Category and SubCategory.

# Results		Messages									
	CustomerKey	CustomerID	FirstName	LastName	Gender	City	JoinDate	StartDate	EndDate	IsCurrent	Birthdate
1	1	C001	Michael	Davis	M	Osborneport	2022-09-25	2026-03-29 22:21:14.000	NULL	1	1996-09-11
2	2	C002	Michael	Miller	M	New Gabrielleport	2020-11-03	2026-03-29 22:21:14.000	NULL	1	1959-08-18
3	3	C003	Carol	Hays	F	Port Allen	2024-02-12	2026-03-29 22:21:14.000	NULL	1	2005-04-19
4	4	C004	Joseph	Ward	M	East Edgarborough	2024-09-09	2026-03-29 22:21:14.000	NULL	1	1992-06-16
5	5	C005	Jamie	Salinas	M	Port Kimberly	2022-02-24	2026-03-29 22:21:14.000	NULL	1	1992-06-18
6	6	C006	Danny	Moore	M	Costaberg	2022-07-09	2026-03-29 22:21:14.000	NULL	1	1986-10-03
7	7	C007	Erin	Walker	M	West Jasmine	2021-10-13	2026-03-29 22:21:14.000	NULL	1	1985-01-11
8	8	C008	Isaiah	Williams	M	New Michaeland	2025-03-04	2026-03-29 22:21:14.000	NULL	1	1960-11-13
9	9	C009	Richard	Pacheco	F	West Lisaville	2025-04-17	2026-03-29 22:21:14.000	NULL	1	2004-08-15
10	10	C010	Catherine	Rice	M	West Carol	2025-07-23	2026-03-29 22:21:14.000	NULL	1	1976-12-12
11	11	C011	Haley	Williams	M	Leburgh	2023-05-12	2026-03-29 22:21:14.000	NULL	1	1997-03-03
12	12	C012	Dale	Perry	M	Lake Dawn	2021-07-16	2026-03-29 22:21:14.000	NULL	1	1989-12-08
13	13	C013	Robert	Anderson	M	Jillianhaven	2023-03-22	2026-03-29 22:21:14.000	NULL	1	1955-01-28
14	14	C014	Brittany	Green	M	East Samantha	2021-11-09	2026-03-29 22:21:14.000	NULL	1	1974-06-25
15	15	C015	Matthew	Molina	M	Quinnville	2022-11-27	2026-03-29 22:21:14.000	NULL	1	1993-12-12
16	16	C016	Miguel	Johnson	M	Murrayborough	2022-02-21	2026-03-29 22:21:14.000	NULL	1	1976-04-27
17	17	C017	Yvonne	Anderson	F	Delaville	2024-09-26	2026-03-29 22:21:14.000	NULL	1	1993-02-11

3. **DimCustomer**: Stores customer details and handles history via SCD Type 2.

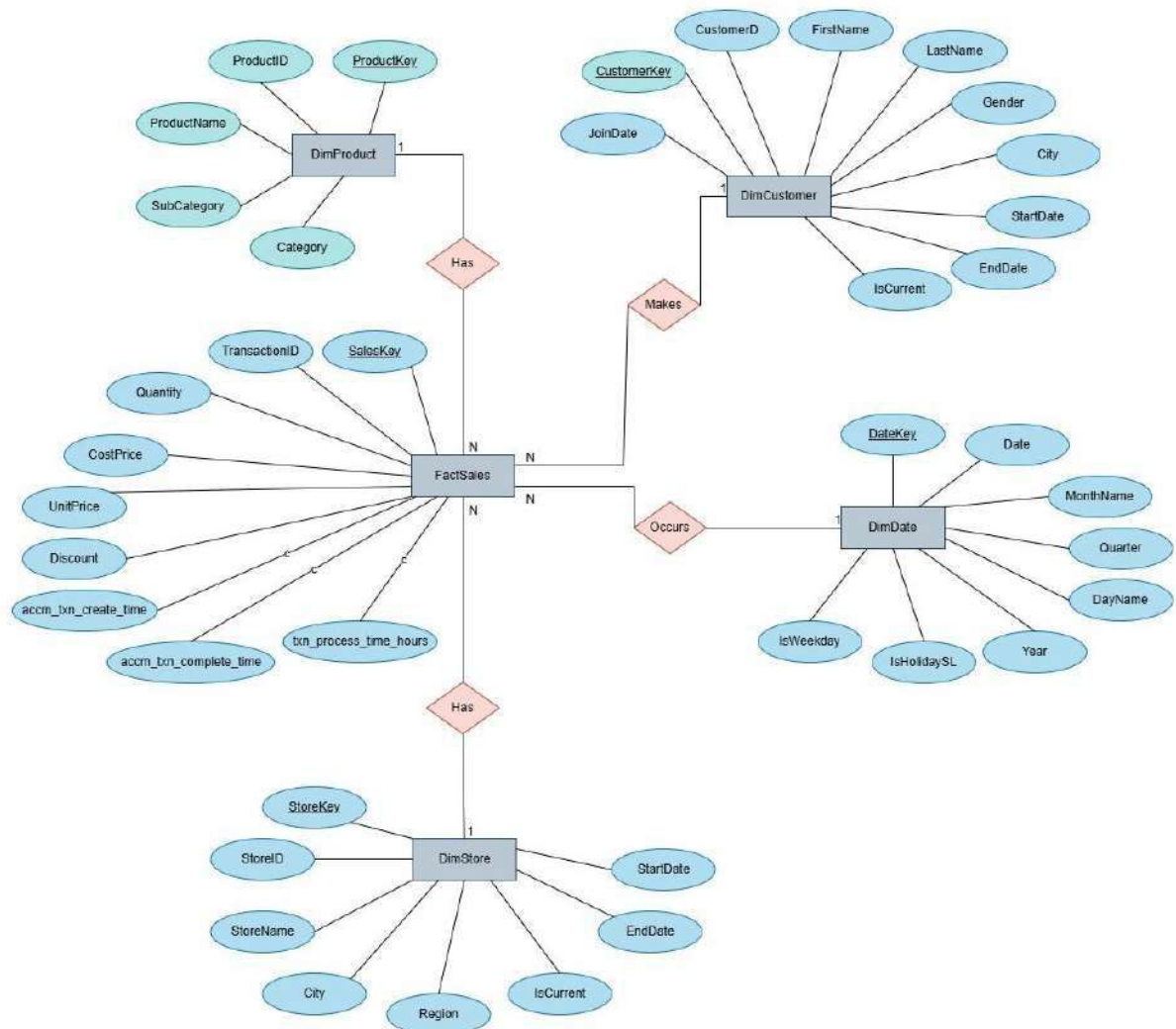
Results		Messages						
	StoreKey	StoreID	StoreName	City	Region	StartDate	EndDate	IsCurrent
1	1	S001	MegaMart Jimenezborough	Jimenezborough	South	2026-03-29 21:14:21.000	NULL	1
2	2	S002	MegaMart Peckmouth	Peckmouth	East	2026-03-29 21:14:21.000	NULL	1
3	3	S003	MegaMart New Michele	New Michele	West	2026-03-29 21:14:21.000	NULL	1
4	4	S004	MegaMart Brianahaven	Brianahaven	North	2026-03-29 21:14:21.000	NULL	1
5	5	S005	MegaMart Johnmouth	Johnmouth	East	2026-03-29 21:14:21.000	NULL	1

4. **DimStore:** Provides geographical context (City/Region).

	DateKey	Date	FullDateUK	FullDateUSA	DayOfMonth
1	19900101	1990-01-01 00:00:00.000	01/01/1990	01/01/1990	1
2	19900102	1990-01-02 00:00:00.000	02/01/1990	01/02/1990	2
3	19900103	1990-01-03 00:00:00.000	03/01/1990	01/03/1990	3
4	19900104	1990-01-04 00:00:00.000	04/01/1990	01/04/1990	4
5	19900105	1990-01-05 00:00:00.000	05/01/1990	01/05/1990	5
6	19900106	1990-01-06 00:00:00.000	06/01/1990	01/06/1990	6
7	19900107	1990-01-07 00:00:00.000	07/01/1990	01/07/1990	7
8	19900108	1990-01-08 00:00:00.000	08/01/1990	01/08/1990	8
9	19900109	1990-01-09 00:00:00.000	09/01/1990	01/09/1990	9
10	19900110	1990-01-10 00:00:00.000	10/01/1990	01/10/1990	10
11	19900111	1990-01-11 00:00:00.000	11/01/1990	01/11/1990	11
12	19900112	1990-01-12 00:00:00.000	12/01/1990	01/12/1990	12
13	19900113	1990-01-13 00:00:00.000	13/01/1990	01/13/1990	13
14	19900114	1990-01-14 00:00:00.000	14/01/1990	01/14/1990	14
15	19900115	1990-01-15 00:00:00.000	15/01/1990	01/15/1990	15

5. **DimDate:** A comprehensive time dimension for trend analysis.

Entity Relationship Diagram (ER)

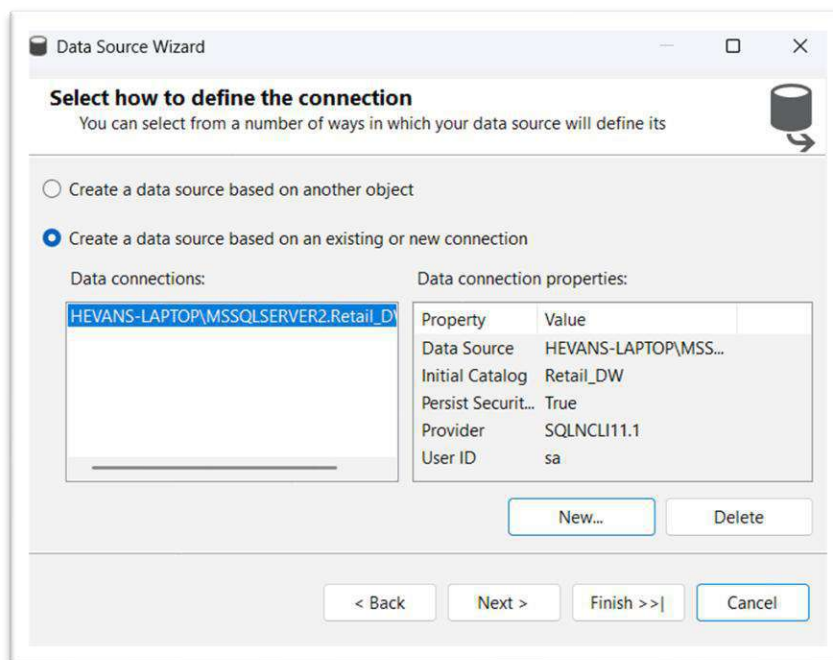


Step 2: SSAS Cube implementation

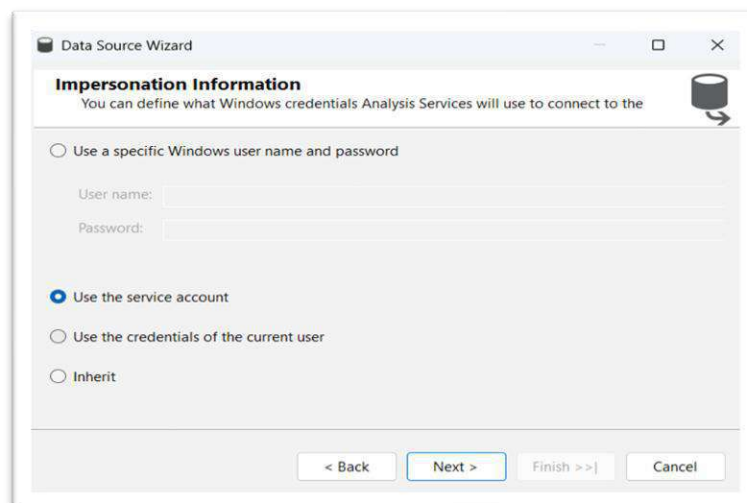
First I created a new **Analysis Services Multidimensional and Data Mining Project** in Visual Studio and renamed it to **Retail_SSAS**.

2.1 Creating a data source

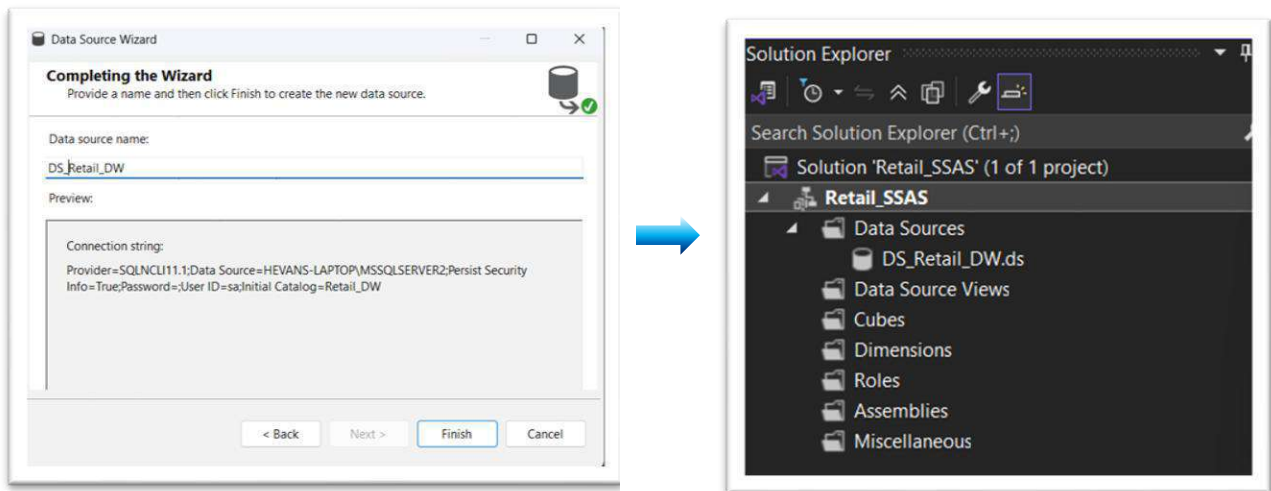
1. I right clickd on “Data Sources” and clicked “New data source” to open **Data Source Wizard**.
2. In the **Select how to define the connection window**, I selected “Create a data source based on an existing or new connection option” and selected the **Retail_DW** database connection in the **Data connections** tab.



3. In the Impersonation Information page, I selected the method “Use the service account” option.

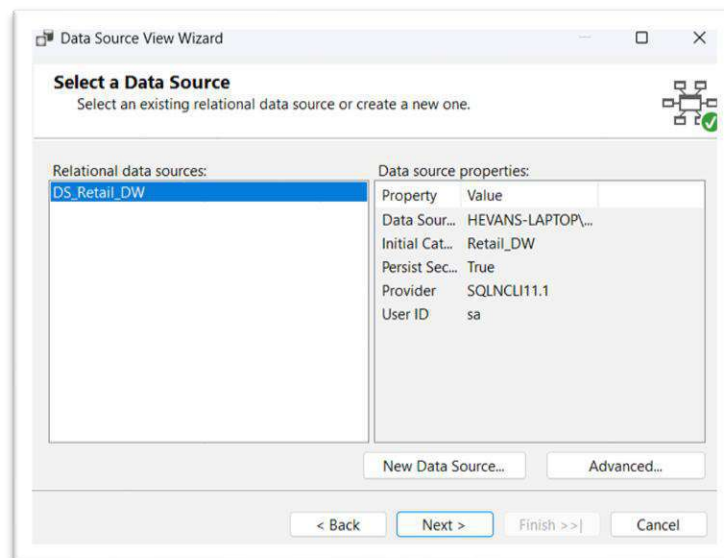


4. After clicking the “Next”, I provided a data source name as **DS_Retail_DW** and clicked “Finish”.

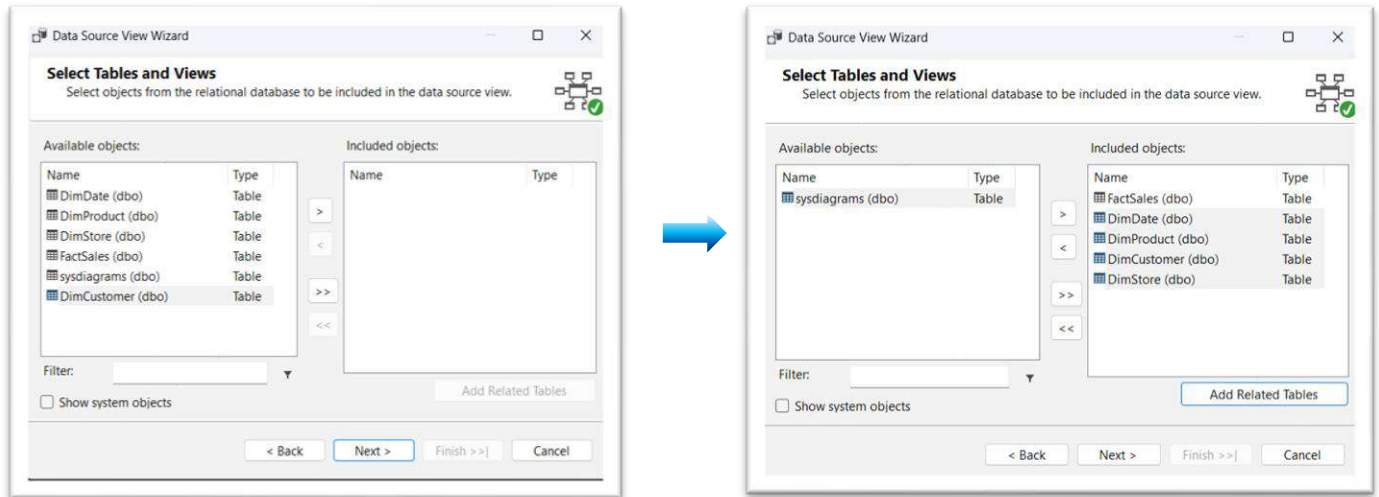


2.2 Creating a Data Source View

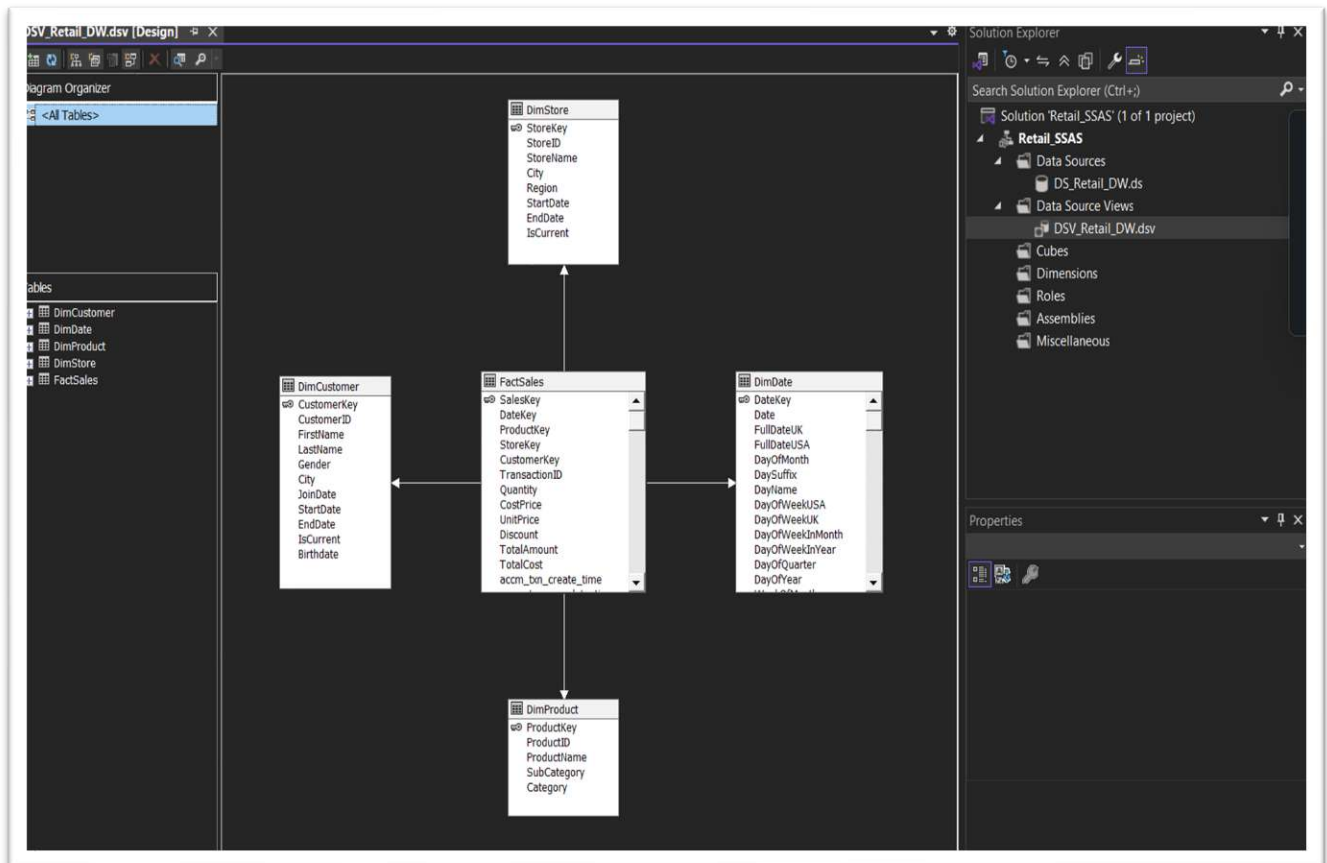
1. First I right clicked on “Data Source Views” and select “New Data Source View”.
2. Next in the **Select a Data Source** page, I selected the data source **DS_Retail_DW** from the **Relational data sources** list and clicked “Next”.



3. After that, in the **Select Tables and Views** page, I first moved **FactSales** table to the **Included objects** from **Available objects** tab using “>” button.
4. Then I clicked “Add related tables” button to add all the dimension tables to the **Included objects**.

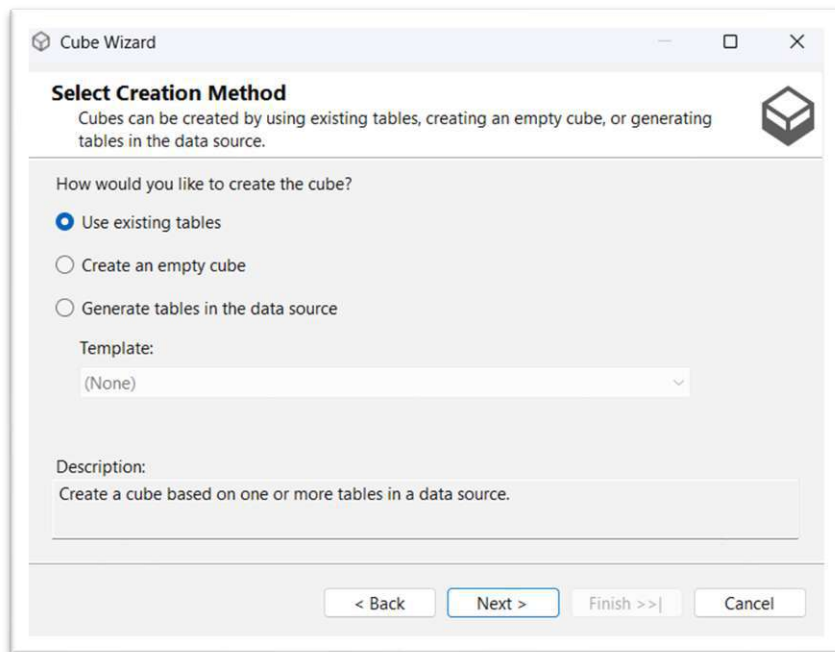


- After clicking “Next” I provided the name **DSV_Retail_DW** for the Data view source and clicked “Finished”. And I viewed it as below.

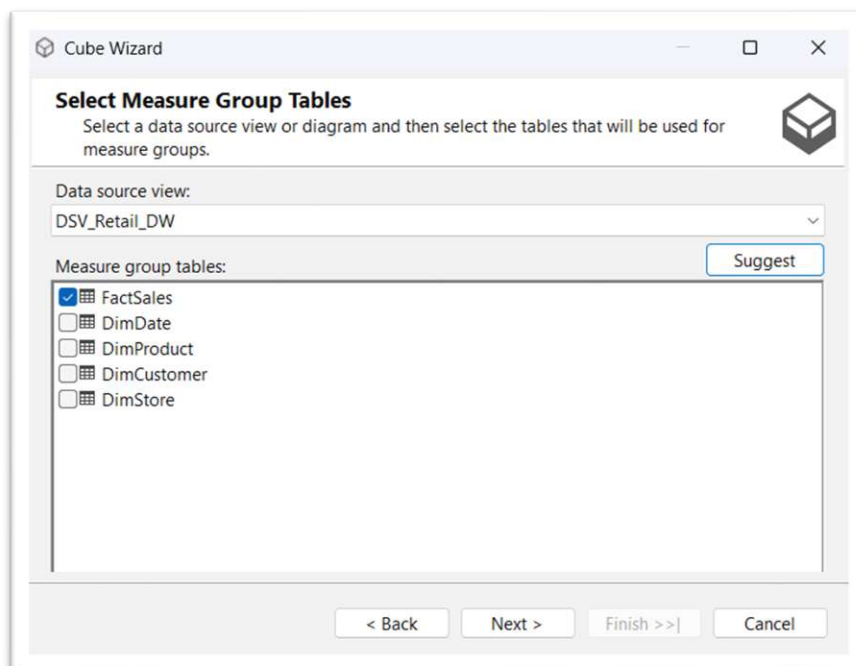


2.3 Creating a SSAS Cube

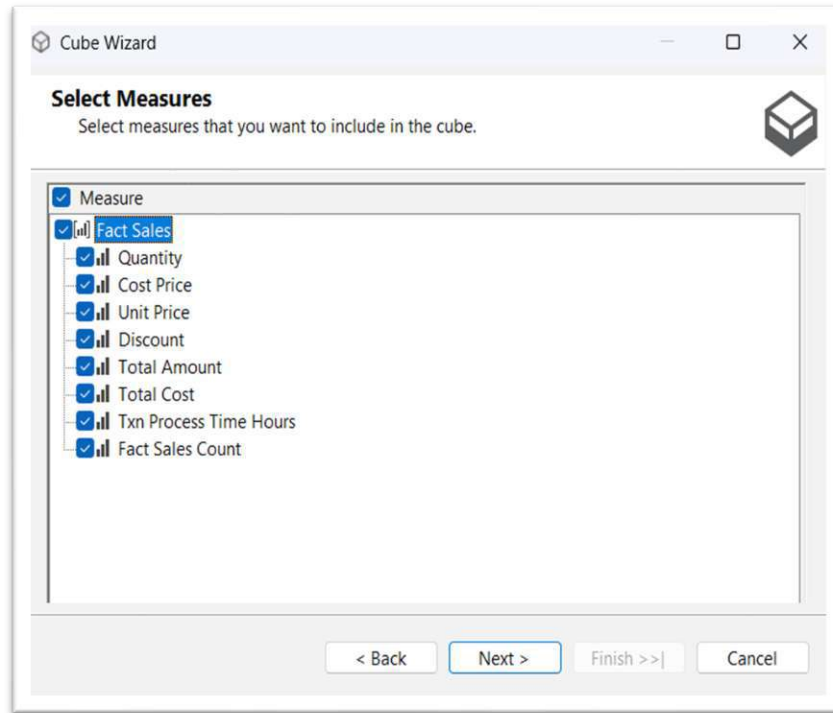
1. I right clicked on “Cubes” and selected “New Cube”
2. In the **Cube Wizard**, on the **Select Creation Page** I selected **Use existing tables**.



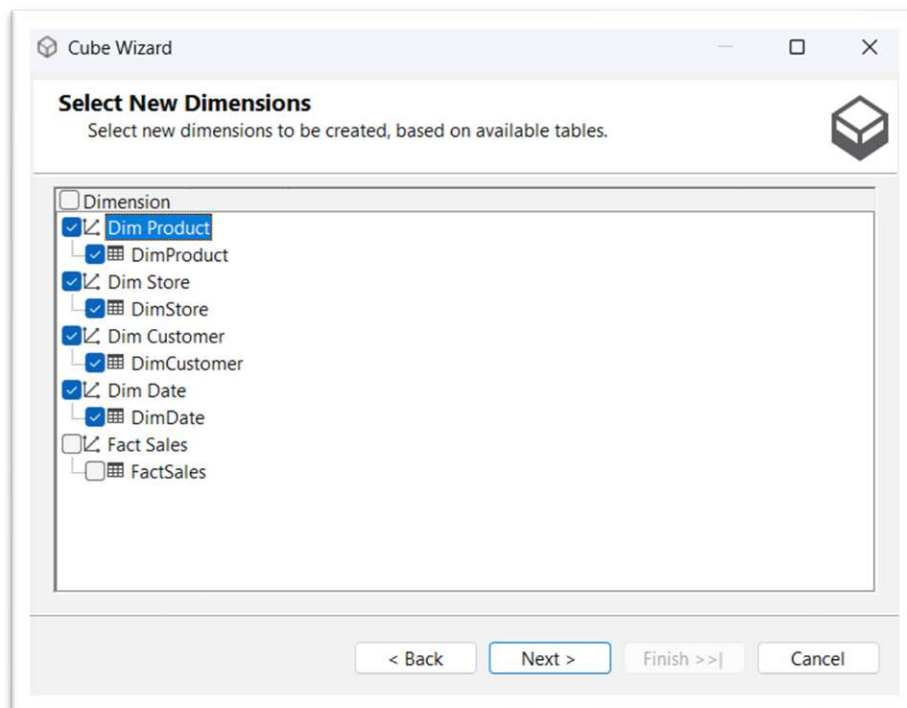
3. In the **Select Measure Group Tables** page, I selected the data source view **DSV_Retail_DW** and selected **FactSales** from **Measure group tables** tab.



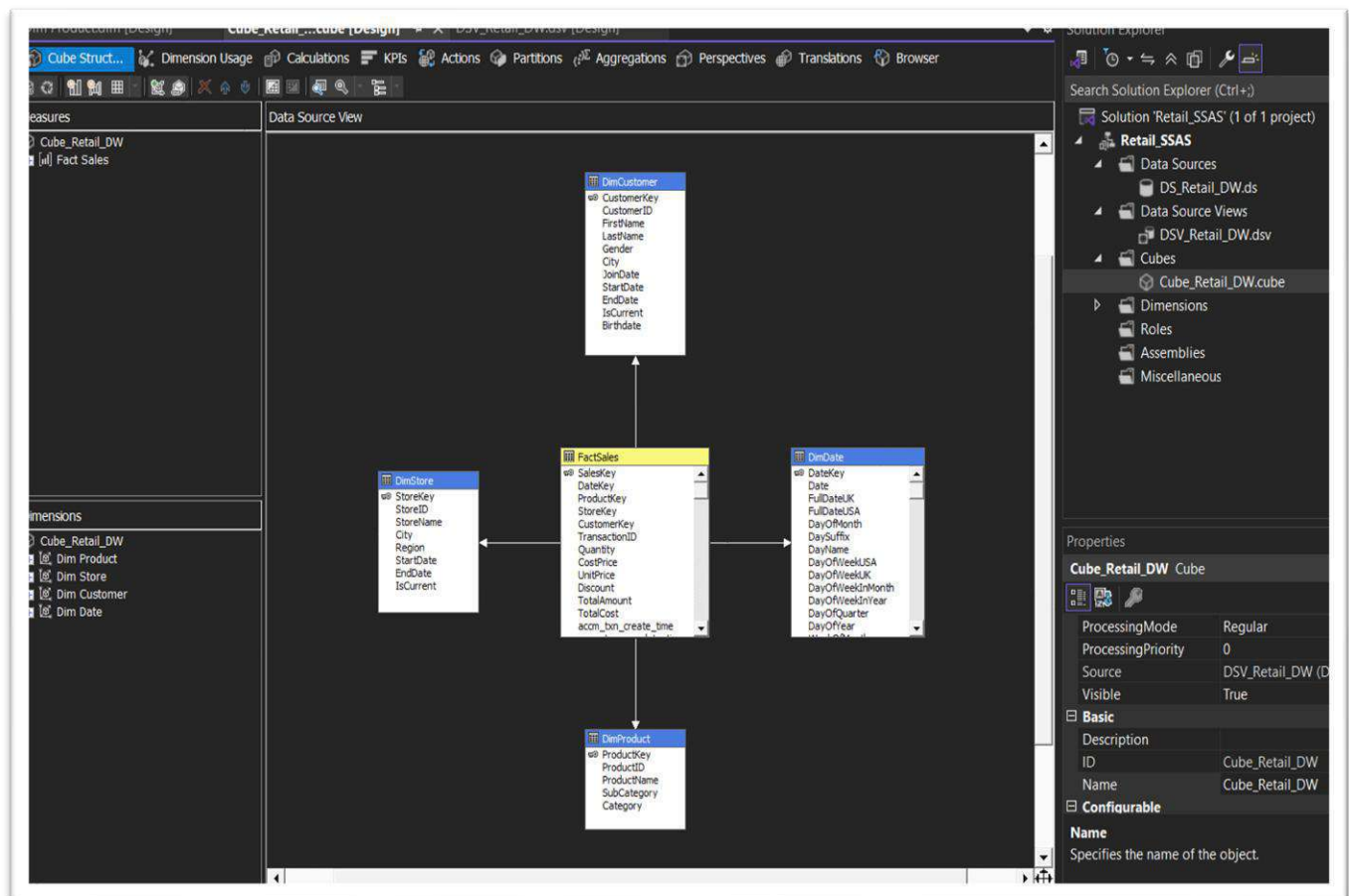
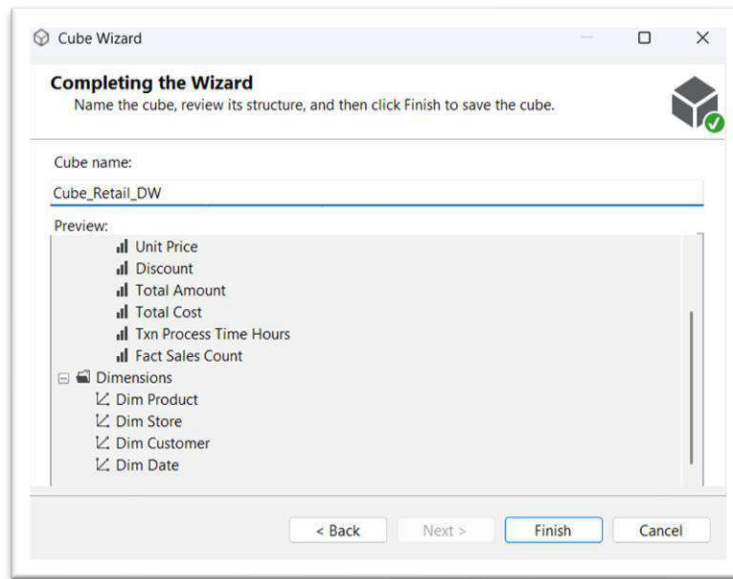
4. In the **Select Measures** page, I selected all the fields and clicked “Next”.



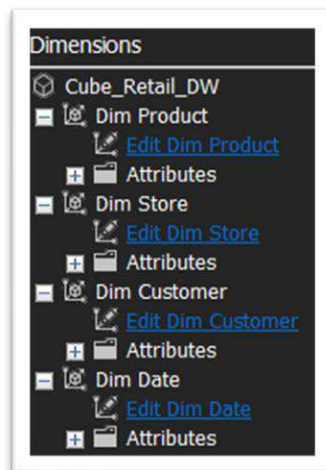
5. In the **Select New Dimensions** page, I selected all the dimension tables and clicked “Next”.



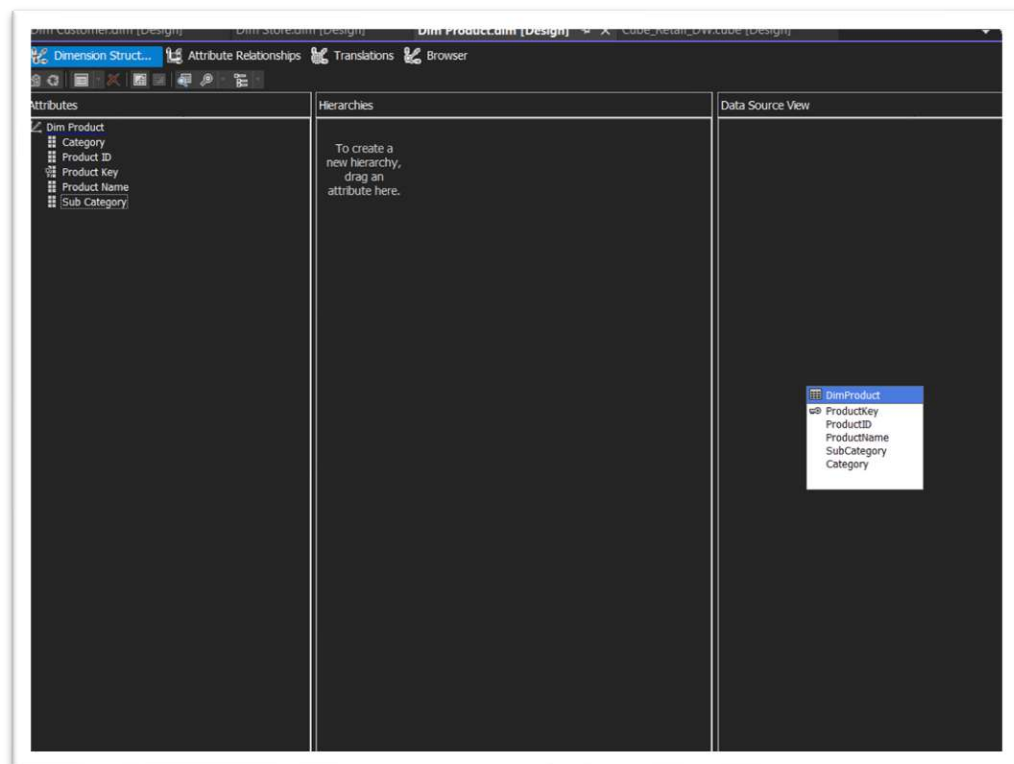
6. I provided the name as **Cube_Retail_DW** for the cube name in the next page and clicked “Next”.



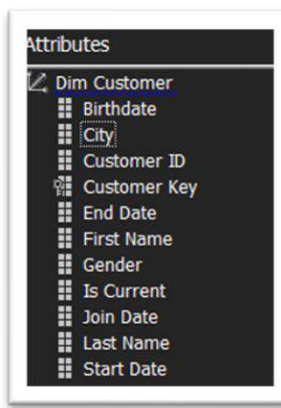
7. In the **Dimensions** section, I expanded “Dimproduct” and then “Attributes”.



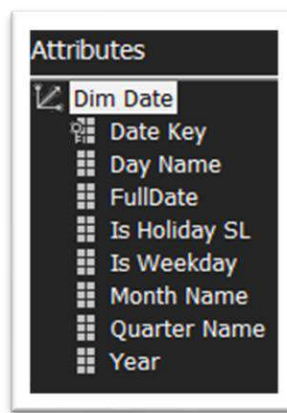
8. Since that only the **Product Key** is listed as an attribute I need to add other attributes.
9. To add rest of the attributes, I clicked on the “Edit Dim Product” and it opened **Dimension editor window** for “DimProduct”
10. In there I selected all the necessary fields except for “ProductKey” in **DimProduct** table in **Data source view** section and added them to the **Attributes** section.



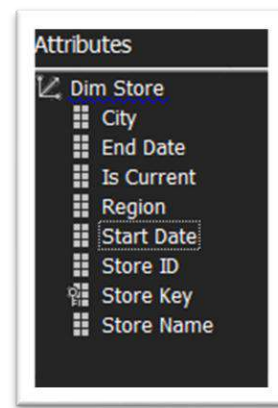
11. After that I saved it.
12. Following the same steps I did the same for DimCustomer, DimDate and DimStore tables.



Dim Customer



Dim Date



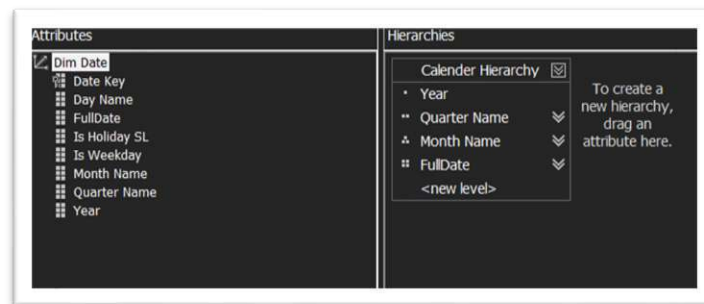
Dim Store

2.4 Creating Hierarchies

Callender Hierarchy

Date → Month → Quarter → Year

1. First I opened “Dim Date.dim” by double clicking on it.
2. Next I dragged and dropped the attributes of **Year, Quarter Name, Month Name** and **FullDate** in to the **Hierarchy** tab.



3. Now in the **Attribute Relationships** tab I created below relationships.
 - Date Key - FullDate (Flexible relationship)
 - FullDate - Month Name (Flexible relationship)
 - Month Name – Quarter Name (Flexible relationship)
 - Quarter Name – Year (Flexible relationship)



- Next I ensured that other attributes are also has relationship with **Date Key**.

Attributes	Attribute Relationships
Date Key	Date Key @ Day Name
Day Name	Date Key @ FullDate
FullDate	Date Key @ Is Holiday SL
Is Holiday SL	Date Key @ Is Weekday
Is Weekday	FullDate @ Month Name
Month Name	Month Name @ Quarter Na...
Quarter Name	Year
Year	

- After that I saved it using “ctrl + S” .

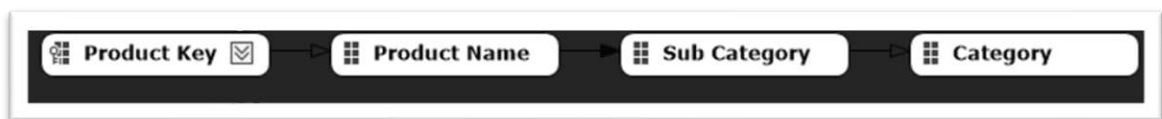
Product Hierarchy

Product Name ➡ Sub Category ➡ Category

- Like I did in the “Calender Hierarchy”, I dragged and dropped **Product Name**, **Sub Category** and **Category** attributes in to the **Hierarchy** tab.

Attributes	Hierarchies
Dim Product	Product Hierarchy
Category	Category
Product ID	Sub Category
Product Key	Product Name
Product Name	<new level>
Sub Category	

- Next I created below relationships between attributes I added.
 - Product Name – Sub Category (Rigid relationship (Because every product has a sub category))
 - Sub Category – Category (Flexible Relationship (Because some sub categories don’t have categories))



- And also I checked the relationships between other attributes as well and saved it.

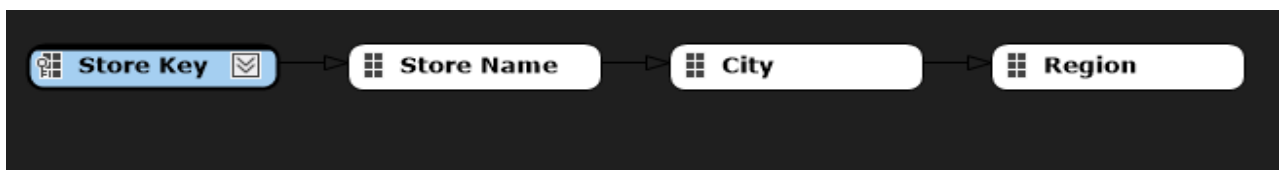
Store Hierarchy

Store Name ➡ City ➡ Region

1. I dragged and dropped **Store Name**, **City** and **Region** attributes in to the **Hierarchy** tab.



2. Next I created below relationships between attributes I added.
 - Store Name – City (Flexible relationship)
 - City – Region (Flexible relationship)

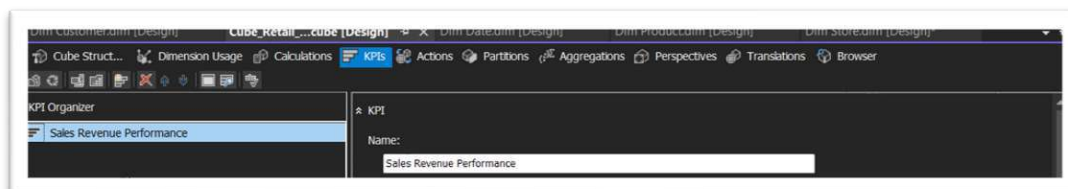


3. And also I checked the relationships between other attributes as well and saved it.

2.5 Creating KPI (Key Performance Indicator)

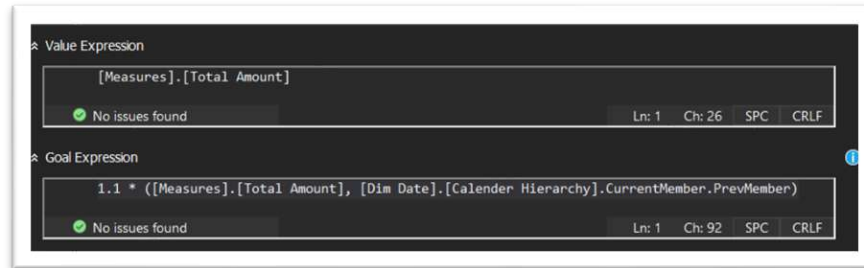
A Key Performance Indicator (KPI) is a quantifiable, time-bound measure used to evaluate the success of an organization, employee, or project in meeting specific strategic objectives. In here I used a KPI to check whether these retail sales met a specific target. I'm going to check that each sales of a period, is 10% higher than the previous period using **Total Amount** and **Calender Hierarchy** .

1. To that, First I created the **Sales Revenue Performance** KPI using the **Cube_Retail_DW.cube** .



2. Next I chose **Fact Sales** as the “Associated measure group” and set the “Value expression” and “Goal expression” as below.

- **Value expression** : [Measures].[Total Amount] (Measuring column used)
- **Goal expression** : 1.1 * ([Measures].[Total Amount], [Dim Date].[Calendar Hierarchy].CurrentMember.PrevMember) (To check the goal of 10% increase in sales)

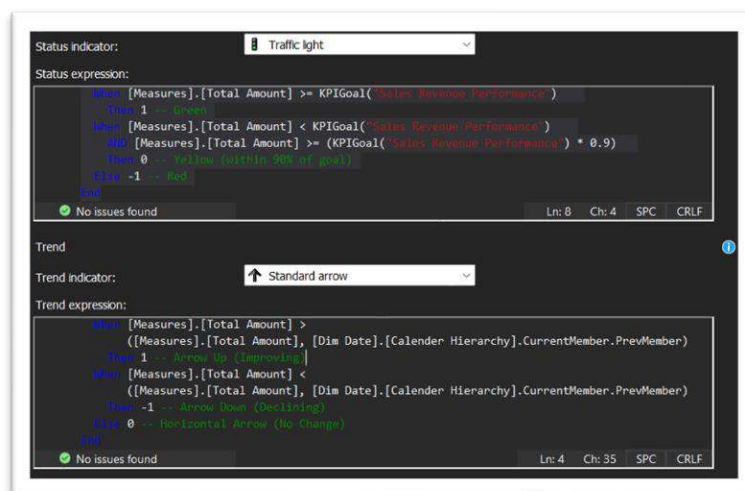


3. And I added **Status** and **Trend** using the below expressions and I chose **Traffic light** as “Status indicator” and **Standard Arrow** as “Trend indicator”. After that I saved it.

- **Status expression** :

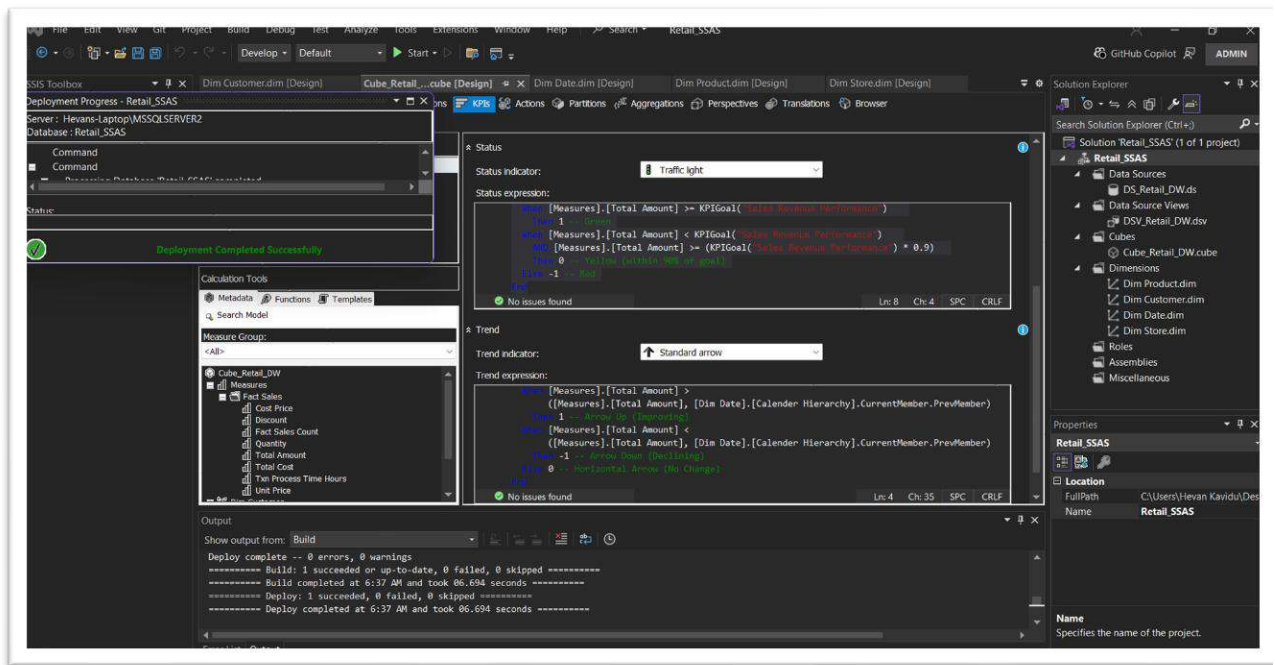
```
Case
  When [Measures].[Total Amount] >= KPIGoal("Sales Revenue Performance")
    Then 1 -- Green(Good goal met)
  When [Measures].[Total Amount] < KPIGoal("Sales Revenue Performance")
    AND [Measures].[Total Amount] >= (KPIGoal("Sales Revenue Performance") * 0.9)
    Then 0 -- Yellow (within 90% of goal)
  Else -1 -- Red(Goal failed)
End
```
- **Trend expression** :

```
Case
  When [Measures].[Total Amount] >
    ([Measures].[Total Amount], [Dim Date].[Calendar Hierarchy].CurrentMember.PrevMember)
    Then 1 -- Arrow Up (Improving)
  When [Measures].[Total Amount] <
    ([Measures].[Total Amount], [Dim Date].[Calendar Hierarchy].CurrentMember.PrevMember)
    Then -1 -- Arrow Down (Declining)
  Else 0 -- Horizontal Arrow (No Change)
End
```

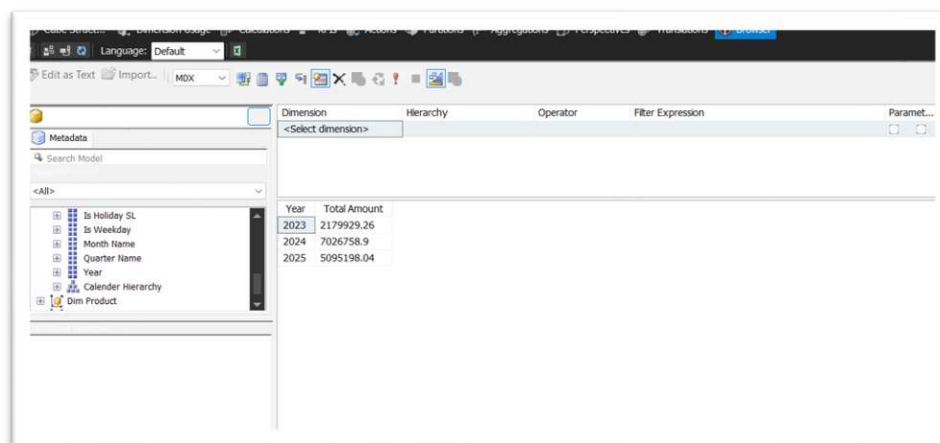


2.6 Deploy the Cube

1. I right clicked on project name “Retail_SSAS” and select deploy .



2. Since the deployment is successful. I checked it using **Browser** tab in “Cub_Retail_DW.cube” window. I dragged dropped attributes in to design area to check it.



3. And I also checked the hierarchies I created.

Year	Quarter Name	Month Name	FullDate	Total Amount
2023	Fourth	December	2023-1...	16651.9
2023	Fourth	December	2023-1...	28575.26
2023	Fourth	December	2023-1...	14011.34
2023	Fourth	December	2023-1...	36868.67
2023	Fourth	December	2023-1...	24621.27
2023	Fourth	December	2023-1...	9699.29
2023	Fourth	December	2023-1...	13541.16
2023	Fourth	December	2023-1...	9034.7
2023	Fourth	December	2023-1...	18254.8
2023	Fourth	December	2023-1...	29627.12
2023	Fourth	December	2023-1...	51545.83
2023	Fourth	December	2023-1...	11114.26
2023	Fourth	December	2023-1...	21219.52
2023	Fourth	December	2023-1...	18245.51

Calendar Hierarchy

Category	Sub Category	Product Name	Total Amount
Electronics	Camera	Above Camera	314677.17
Electronics	Camera	Like Camera	482608.150...
Electronics	Camera	Understand C...	333519.35
Fashion	Headphones	Either Headp...	67747.3999...
Fashion	Headphones	Find Headpho...	341309.13
Fashion	Headphones	Piece Headph...	497149.18
Groceries	Laptop	Chair Laptop	429824.15
Groceries	Laptop	Everything La...	134386.01
Groceries	Laptop	Traditional La...	521030.72
NA	Accessories	Beat Accessor...	498224.37
NA	Accessories	Four Accesso...	471227.94
NA	Accessories	House Access...	432995.57
NA	Accessories	Least Accesso...	297321.49
NA	Accessories	Simple Access...	82198.61

Product Hierarchy

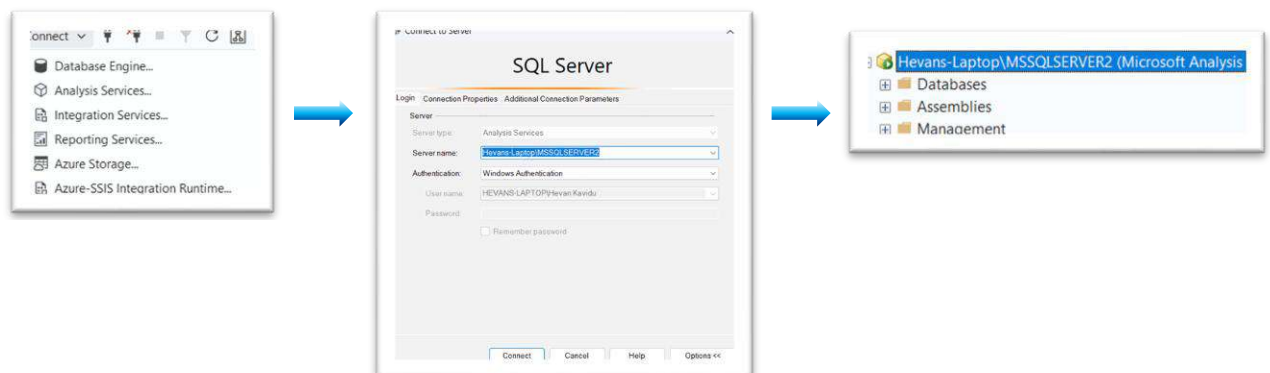
Region	City	Store Name	Total Amount
East	Jo...	MegaMart J...	2847044.49
East	Pe...	MegaMart ...	2753629.18
North	Bri...	MegaMart ...	2900697.6
South	Jim...	MegaMart J...	2880111.26...
West	Ne...	MegaMart ...	2920403.66

Geographical Hierarchy

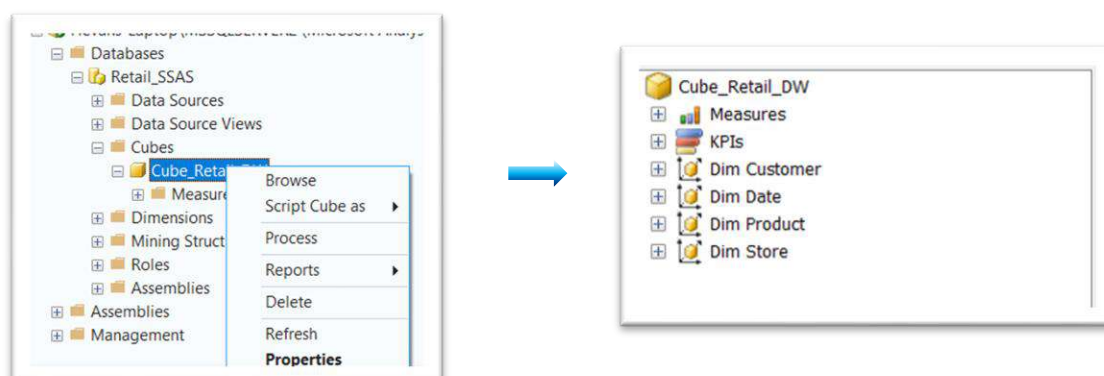
4. Next I checked the **Sales Revenue Performance KPI**.

Year	Quarter Name	Month Name	FullDate	Sales Revenue Performance V...	Sales Revenue Performance G...	Sales Revenue Performance S...	Sales Revenue Performance T...
2023	Third	September	2023-0...	18327	38452.26	-1	-1
2023	Third	September	2023-0...	21423.86	20159.7	1	1
2023	Third	September	2023-0...	21715.33	23566.246	0	1
2023	Third	September	2023-0...	16927	23886.863	-1	-1
2023	Third	September	2023-0...	34443.86	18619.7	1	1
2023	Third	September	2023-0...	17559.14	37888.246	-1	-1
2023	Third	September	2023-0...	37697.12	19315.054	1	1
2023	Third	September	2023-0...	13609.6	41466.832	-1	-1
2023	Third	September	2023-0...	13936.06	14970.56	0	1
2023	Third	September	2023-0...	2187.13	15329.666	-1	-1
2023	Third	September	2023-0...	15967.87	2405.843	1	1
2023	Third	September	2023-0...	20715.22	17564.657	1	1
2023	Third	September	2023-0...	10208.26	22786.742	-1	-1

5. After checking all of them next I checked deployment in SQL Server Management Studio. By connecting to “Analysis services”.



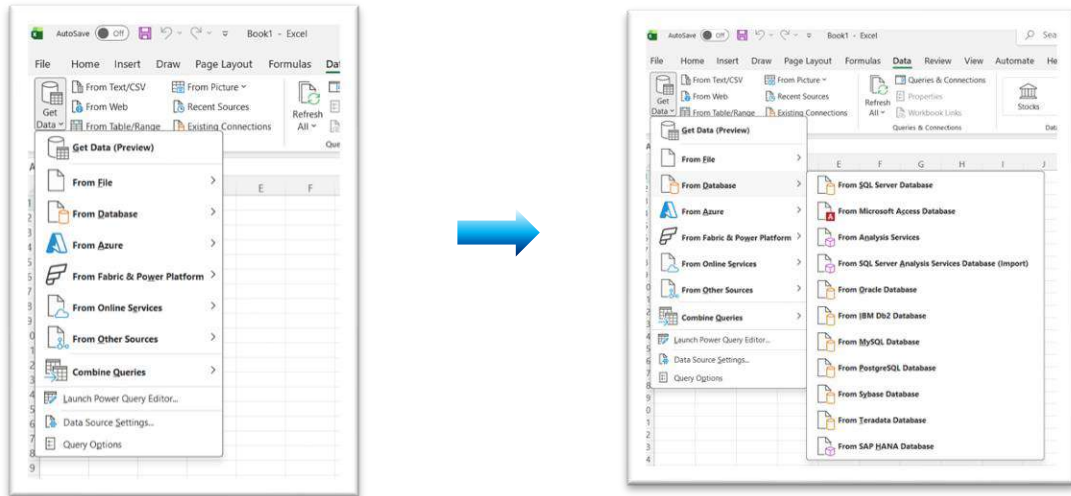
6. By expanding “Databases” I found the cube and right clicked on it and select “Browse” and it opened the design window like in **Visual Studio**.



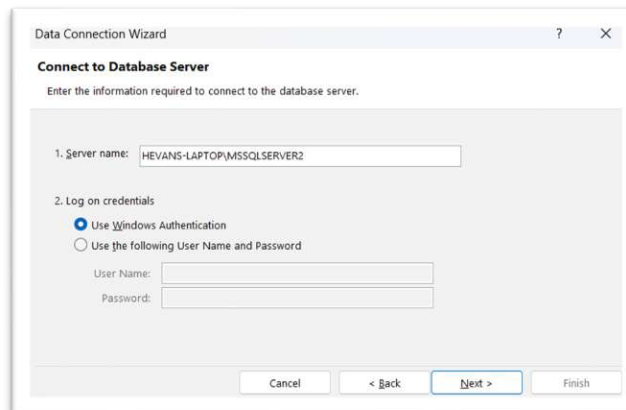
Step 3: Demonstration of OLAP operations

Connecting Excel workbook to cube

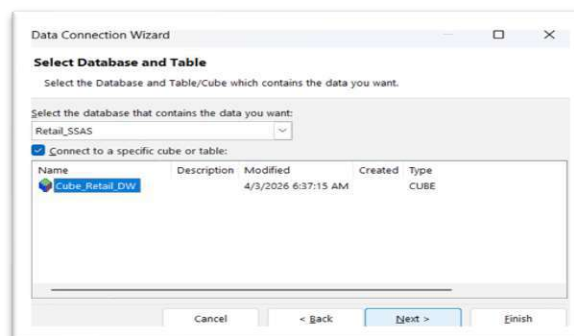
1. First I opened a blank excel workbook.
2. Next I selected “Get data”, clicked on “From Database” and clicked on “From Analysis Services”.



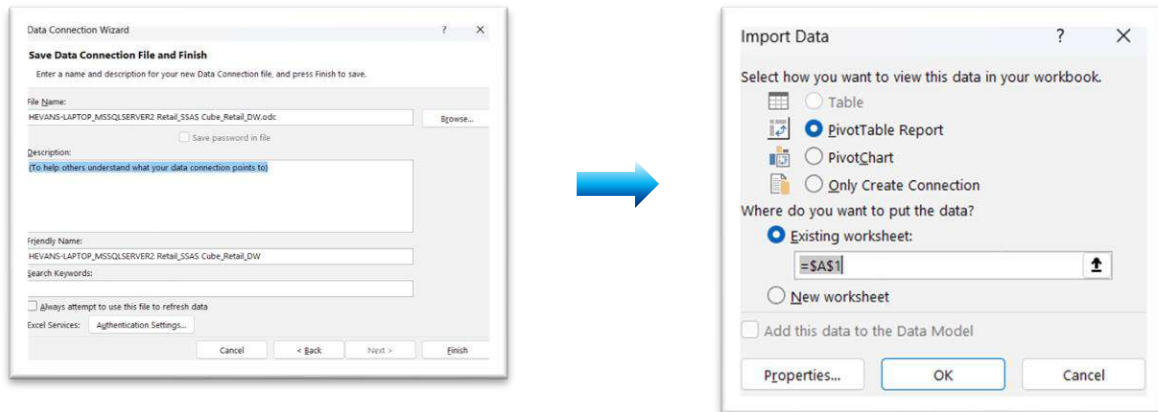
3. In the **Data Connection Wizard** I selected the server as **HEVANS-LAPTOP\MSSQLSERVER2** And clicked “Next”



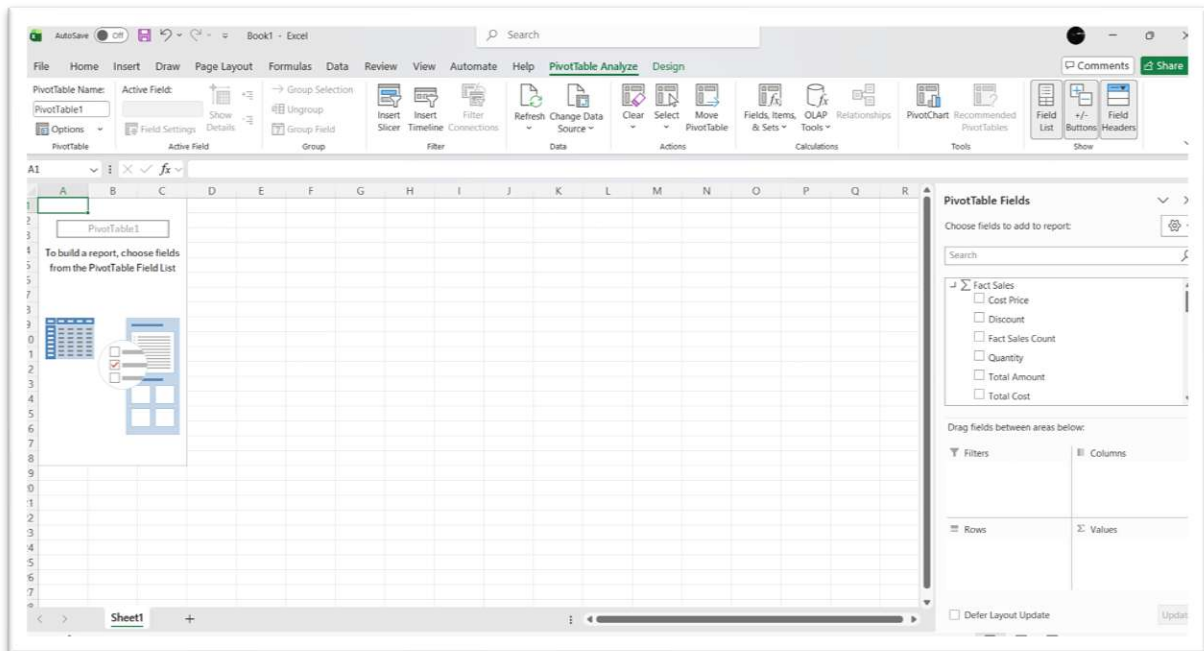
4. In the **Select Database and Table** tab I selected **Retail_SSAS** as the database and **Cube_Retail_DW** as my cube and clicked “Next”.



5. In the **Save data connection File and Finish** tab I clicked “Finished” and in the **Import data** pop up I selected “PivotTable report” to finish configuration.



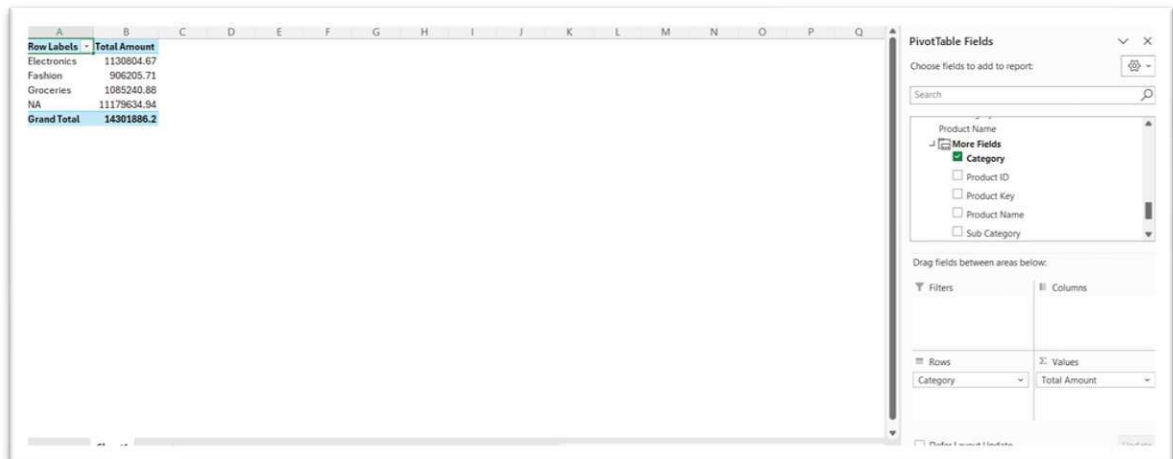
6. Now it will open a new Excel sheet with a **Pivot Table**.



Slice Operation

In here I'm going to analyze "How sales performance varies with **Product Categories** in a specific time period". By selecting a single year from the **Year** dimension, I have 'sliced' the data cube to isolate sales performance for that specific period.

1. In the **PivotTable Fields** I dragged and dropped **Product Category** from **Dim Product** table to "Rows" and **Total Amount** from **Fact Sales** table to "Values". By doing that it created a pivot table.



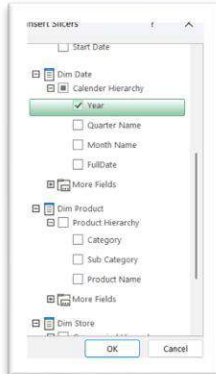
The screenshot shows a PivotTable in Excel with the following data:

Row Labels	Total Amount
Electronics	1130804.67
Fashion	906205.71
Groceries	1085240.88
NA	11179634.94
Grand Total	14301886.2

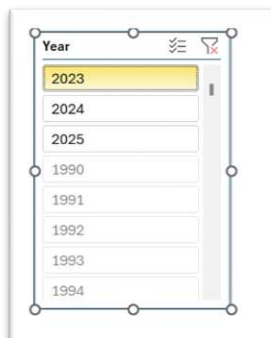
The PivotTable Fields task pane on the right shows the following configuration:

- Choose fields to add to report: Product Name, **Category** (checked), Product ID, Product Key, Product Name, Sub Category.
- Drag fields between areas below:
 - Filters: (empty)
 - Columns: (empty)
 - Rows: Category
 - Values: Total Amount

2. Next on the top ribbon I selected "PivotTable Analyze" and selected "Insert Slicer". It opened a **Insert Slicers** tab.



3. In there I selected **Year** from **Dim Date** as the slicer and clicked "Ok". And it opened a **Year Slicer** like below.



- By selecting a **Year** from there I analyzed how in each year Sales performance changed from category wise.

Row Labels	Total Amount
Electronics	
Camera	
Above Camera	66574.36
Like Camera	87868.67
Understand Camera	70653.95
Fashion	185596.89
Groceries	218966.39
NA	1630239.16
Grand Total	2179929.36

2023

Row Labels	Total Amount
Electronics	
Camera	
Above Camera	22410.89
Like Camera	261095.49
Understand Camera	189055.74
Fashion	450399.14
Groceries	528822.9
NA	5474974.74
Grand Total	7026758.9

2024

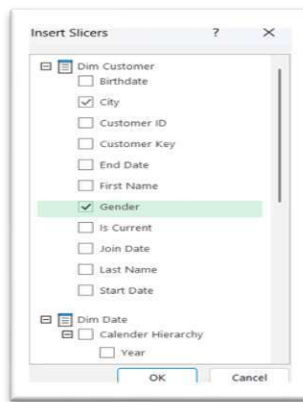
Row Labels	Total Amount
Electronics	
Camera	
Above Camera	125691.92
Like Camera	133643.99
Understand Camera	73779.62
Fashion	350209.88
Groceries	337451.59
NA	4074421.04
Grand Total	5095198.04

2025

Dice Operation

A "Dice" is a multi-dimensional filter. It defines a sub-cube by picking specific values from two or more dimensions. The Dice operation was performed by applying filters across three dimensions (Date, Customer city, and Customer gender). This defines a specific sub-cube of the Retail_DW for deep-dive analysis. Using this we can analyze “How sales performance variates according to **Product Categories** in a specific time period, city and gender” .

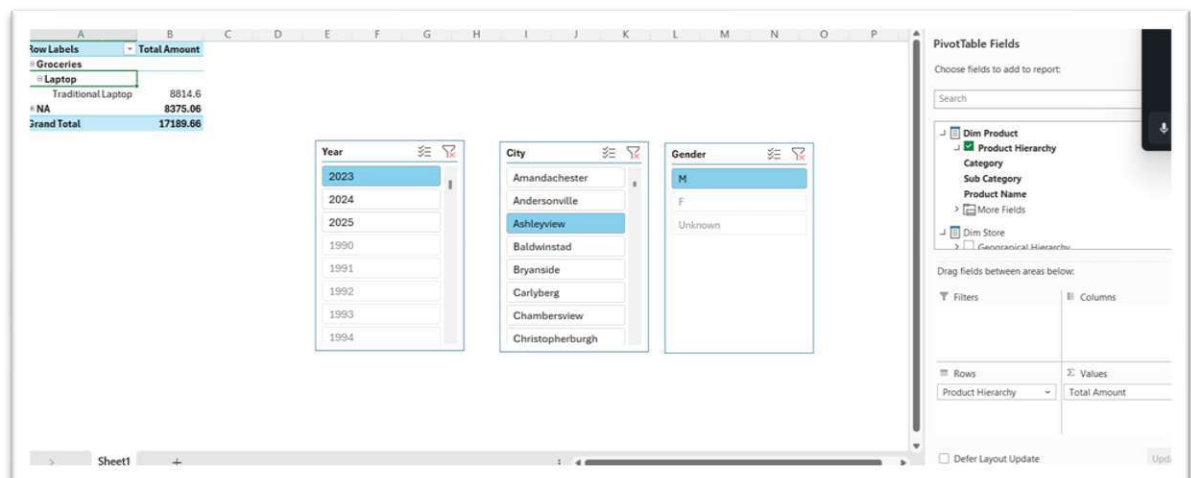
- I used previous slice operation to perform dice operation by adding two more slicers.
- Again I clicked on “Insert Slicer” to add **Gender** and **City** from **Dim Customer** table.



3. After selecting “Gender” and “City” I clicked “Ok” .
4. Next I analyzed data using the slicers.

Example : Company want to see “How much sales made for “Traditional Laptops” by “Male” customers who lives in city “Ashleyview” in year “2023” ?”. Company wish to open a new “Traditional laptop” store in “Ashleyview” , so he can decide whether it is good decision or not based on the result driven below.

- Set year to “2023”
- Set city to “Ashleyview”
- Set gender to “M”
- Expand “Groceries” and next “Laptop” to see “Traditional Laptop”.



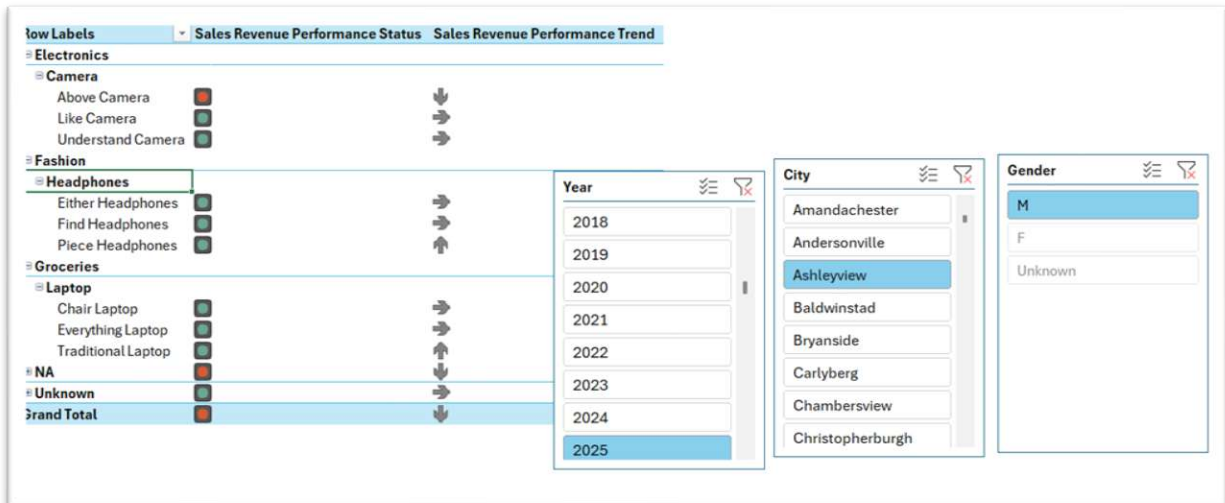
Dice operation using KPI

Since I integrated the **Sales Revenue Performance KPI** with status indicators and trend indicators. I can use them into the Pivot Table to monitor performance visually. I then performed a Dice operation by filtering the data. The SSAS Cube calculates the KPI logic server-side, the Status Icons automatically update as I change the Slicer selections (Dicing). This demonstrates a fully integrated BI solution where the measures, hierarchies, and KPIs work in unison.

1. For the above Excel workbook I used for dicing I dragged and dropped **Status** and **Trend** from “Sales Revenue Performance” KPI.

- Next as you can see in below we can analyze the data by visual representation. In the KPI Goal is set to check whether current period achieved more than 10% growth than last period.

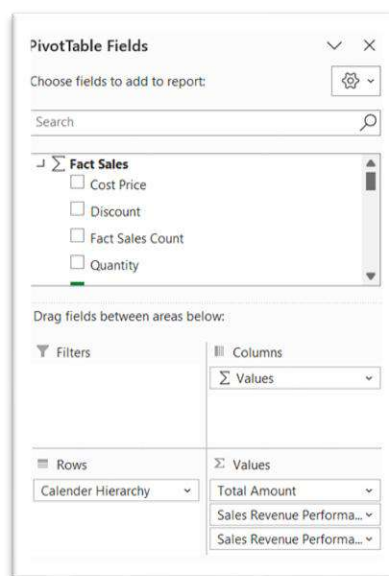
- Red** : Sales performance below the 90% of the goal
- Yellow** : Sales performance achieved 90% of the goal
- Green** : Sales performance achieved the goal



Drill-Down Operation

Drill-Down is moving from a high-level summary (e.g., Total Amount in a Year) to a more detailed level (e.g., Sales in each Month of that Year) to find specific trends. In here I used the **Calender Hierarchy** to drill-down sales from **Year** to **Date**.

- First I added **Calender Hierarchy** to the “Rows” and **Value, Status** and **Trend** from “Sales Revenue Performance” KPI.



2. Next followed below steps to drill-down from year to date.

15	* 2023	2179929.26	
16	* 2024	7026758.9	
17	* 2025	5095198.04	



2023			
	* First		
	* Fourth	1798937.49	
	* Second		
	* Third	380991.77	
	* 2024	7026758.9	
	* 2025	5095198.04	

2. Clicked on (+) to expand Quarters

1. Clicked on (+) on Quarter Name to expand Months

2023			
	* First		
	* Fourth		
	* December	653300.37	
	* November		
	2023-11-01 00:00:00.000	20298	
	2023-11-02 00:00:00.000	6084.13	
	2023-11-03 00:00:00.000	21561.69	
	2023-11-04 00:00:00.000	18940.08	
	2023-11-05 00:00:00.000	30767.78	
	2023-11-06 00:00:00.000	25618.76	
	2023-11-07 00:00:00.000	22203.32	
	2023-11-08 00:00:00.000	6050.08	
	2023-11-09 00:00:00.000	13559.67	
	2023-11-10 00:00:00.000	29416.51	
	2023-11-11 00:00:00.000	12985.15	
	2023-11-12 00:00:00.000	18632.8	
	2023-11-13 00:00:00.000	17380.5	
	2023-11-14 00:00:00.000	22005.1	
	2023-11-15 00:00:00.000	22683.42	
	2023-11-16 00:00:00.000	7118.18	
	2023-11-17 00:00:00.000	33207.56	
	2023-11-18 00:00:00.000	38206.03	
	2023-11-19 00:00:00.000	40484.56	
	2023-11-20 00:00:00.000	14719.4	



2023			
	* First		
	* Fourth		
	* December	653300.37	
	* November	584406.56	
	* October	561230.56	
	* Second		
	* Third	380991.77	
	* 2024	7026758.9	
	* 2025	5095198.04	



3. Now we can analyze sales performance date wise in months.

4. Clicked on (+) on Month Name to expand Dates

Roll-Up Operation

In roll-up operation we moving from detailed data back to a summarized view (e.g., collapsing dates, months, quarters back into a single Year figure) to see the whole summarized analysis.

1. I went top to bottom in my **Calendar Hierarchy** in drill-down operation to analyze data. But here I collapsed the expanded columns in drill-down operation following below steps.

2023			
	* First		
	* Fourth		
	* December	653300.37	
	* November		
	2023-11-01 00:00:00.000	20298	
	2023-11-02 00:00:00.000	6084.13	
	2023-11-03 00:00:00.000	21561.69	
	2023-11-04 00:00:00.000	18940.08	
	2023-11-05 00:00:00.000	30767.78	
	2023-11-06 00:00:00.000	25618.76	
	2023-11-07 00:00:00.000	22203.32	
	2023-11-08 00:00:00.000	6050.08	
	2023-11-09 00:00:00.000	13559.67	
	2023-11-10 00:00:00.000	29416.51	
	2023-11-11 00:00:00.000	12985.15	
	2023-11-12 00:00:00.000	18632.8	
	2023-11-13 00:00:00.000	17380.5	
	2023-11-14 00:00:00.000	22005.1	
	2023-11-15 00:00:00.000	22683.42	
	2023-11-16 00:00:00.000	7118.18	
	2023-11-17 00:00:00.000	33207.56	
	2023-11-18 00:00:00.000	38206.03	
	2023-11-19 00:00:00.000	40484.56	
	2023-11-20 00:00:00.000	14719.4	



2023			
	* First		
	* Fourth		
	* December	653300.37	
	* November	584406.56	
	* October	561230.56	
	* Second		
	* Third	380991.77	
	* 2024	7026758.9	
	* 2025	5095198.04	
	* 2026		

1. I clicked on (-) on Quarter name to collapse month



2. I clicked on (-) on Month name to collapse dates

2023	2179929.26
2024	7026758.9
2025	5095198.04
2026	



2023	
First	
Fourth	1798937.49
Second	
Third	380991.77
2024	7026758.9
2025	5095198.04
2026	

3. In there I can analyze data by year wise(Summary)

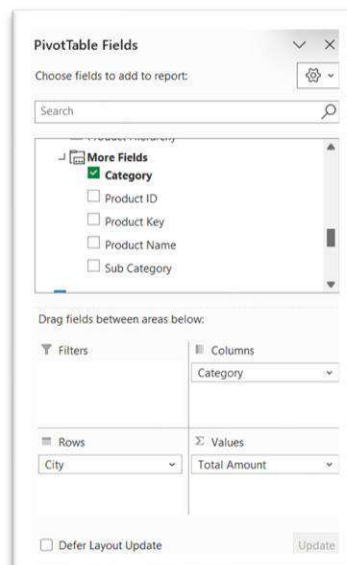
4. I clicked on (-) on year to collapse quarters

Pivot Operation

Pivot operation is about changing the perspective of the analysis instantly to answer different business questions. "Pivoting" means rotating the data axes to view the "cube" from a new angle. Here is how to do it descriptively for your assignment.

Example : First I want to look at “Which store cities are buying which product categories most ?” . Assume that after I want to know “Which product categories are most popular in which city ?”. So I rotate the axis to look at it.

1. In my PivotTable Fields, I dragged and dropped **City** from “Dim Customer” into the **Rows** area.
2. Next I dragged and dropped **Product Category** from “ Dim Product” into the **Columns** area.
3. After that I dragged and dropped **Total Amount** from “Fact Sales” into the **Values** area.
(In here I currently comparing City performance side-by-side across categories.)

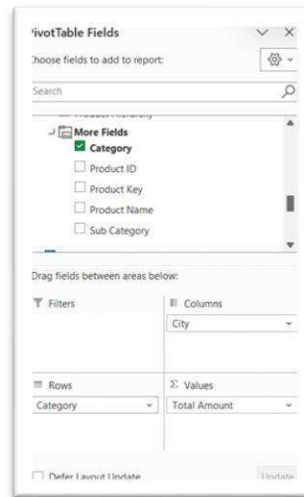


4. After that the table looked like this.

A	B	C	D	E	F
Total Amount	Column Labels				
Row Labels	Electronics	Fashion	Groceries	NA	Grand Total
Brianahaven	224814.02	186089.71	207682.58	2282111.29	2900697.6
Jimenezborough	261235.66	189349.42	279824.22	2149701.97	2880111.27
Johnmouth	256332.02	171886.02	141024.98	2277801.47	2847044.49
New Michele	154870.11	168103.4	225846.42	2371583.73	2920403.66
Peckmouth	233552.86	190777.16	230862.68	2098436.48	2753629.18
Grand Total	1130804.67	906205.71	1085240.88	11179634.94	14301886.2

5. Now to perform the pivot,

- I clicked and dragged **City** from the **Rows** box down to the **Columns** box.
- Simultaneously, I dragged **Product Category** from the **Columns** box into the **Rows** box.



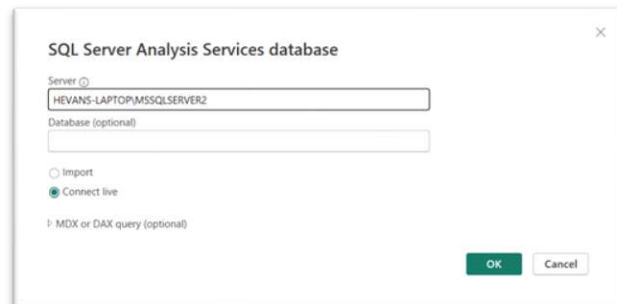
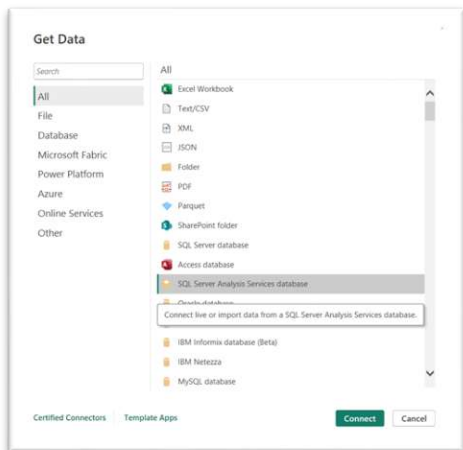
6. Now the table looked like this.

Total Amount	Column Labels					
Row Labels	Brianahaven	Jimenezborough	Johnmouth	New Michele	Peckmouth	Grand Total
Electronics	224814.02	261235.66	256332.02	154870.11	233552.86	1130804.67
Fashion	186089.71	189349.42	171886.02	168103.4	190777.16	906205.71
Groceries	207682.58	279824.22	141024.98	225846.42	230862.68	1085240.88
NA	2282111.29	2149701.97	2277801.47	2371583.73	2098436.48	11179634.94
Grand Total	2900697.6	2880111.27	2847044.49	2920403.66	2753629.18	14301886.2

- The table flipped. Now vertical list is now **Categories** and horizontal headers are **Cities** . By using this we can answer the above example question of “Which product categories are most popular in which city ?”

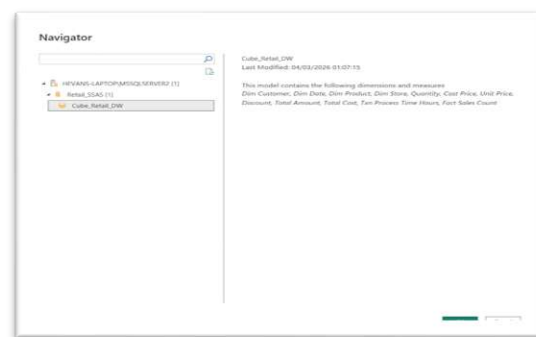
Step 4: PowerBI Reports

First I made the connection between **Power BI desktop** and **SSAS Cube** following below steps.

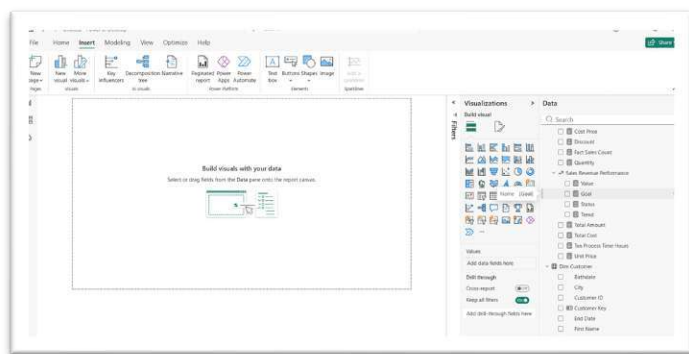


2. Next I entered my server name and clicked on “Ok”

1. First I clicked on “Get data from other sources” and in there I selected **SQL Server Analysis Services Database** and clicked “Connect”



3. Next I selected my cube and clicked “Ok”

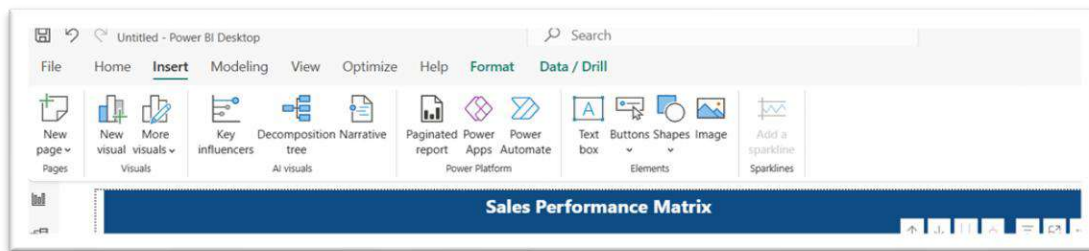


4. Next it will open a new page

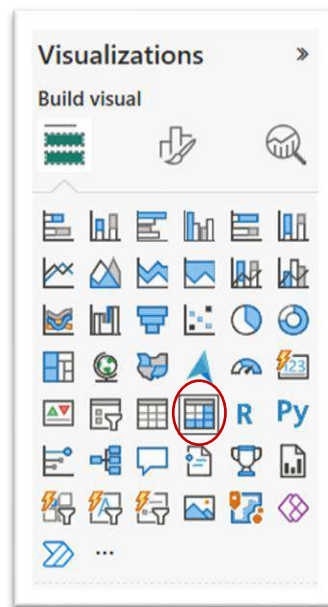
Report 1 :

Report 1 features an advanced Matrix visual that acts as a multi-dimensional pivot table. It integrates the SSAS 'Calendar Hierarchy' on the vertical axis and 'Product Categories' on the horizontal axis. And also it uses KPI to analyze the trend and goal achievement.

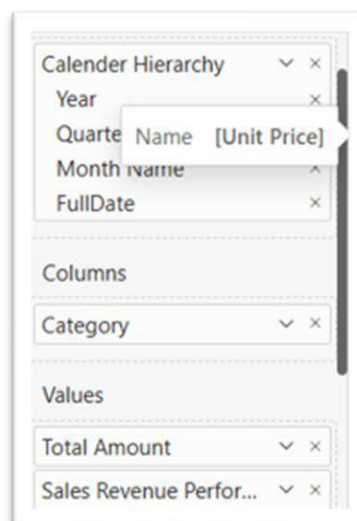
1. First I created a text box using the “Insert” tab in power bi to add a header to the report.



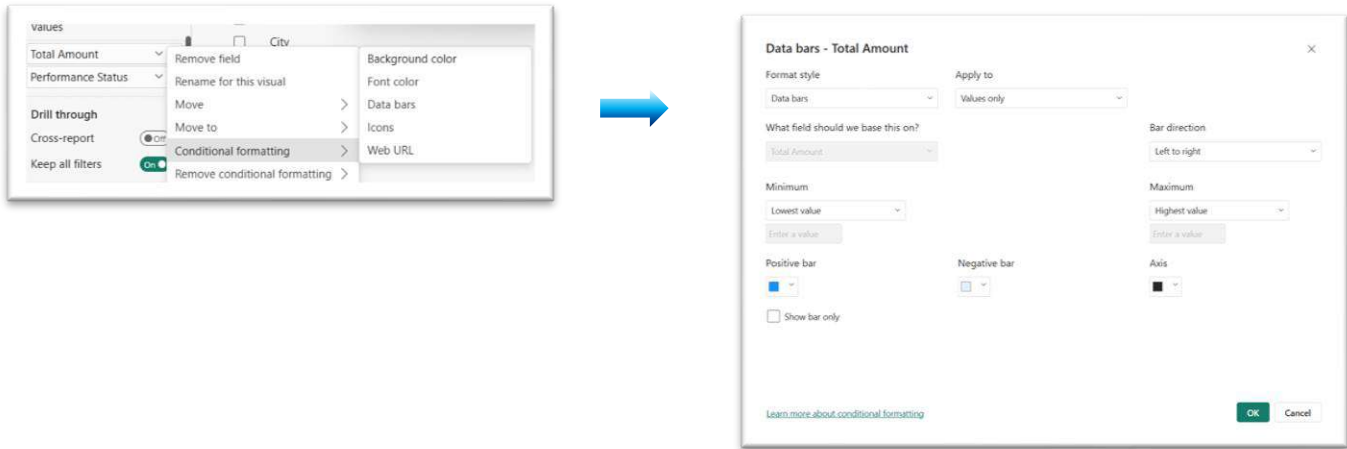
2. Next I selected a **Matrix** from the “Visualizations” tab. And clicked on it to add it to the page.



3. After that I added **Calender Hierarchy** in to the “Rows”, **Category** from the “Dim Product” to the “Columns” and **Total Amount** and **KPI status** in to the “Values”.



4. And next to display data more visually I added data bars to the **Total Amount** column.



Changed the bar direction left to right

5. After click on “Ok” final **Report 1** view is as below.

Sales Performance Matrix

↑

↓

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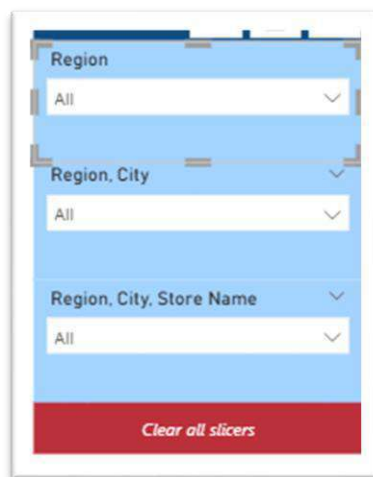
...

Category Year	Electronics		Fashion		Groceries		NA		Unknown	
	Total Amount	Performance Status	Total Amount	Performance Status	Total Amount	Performance Status	Total Amount	Performance Status	Total Amount	Performance Status
2023	225,127.02	🟢	105,596.69	🟢	218,966.39	🟢	1,630,239.16	🟢		🟢
First		🟢		🟢		🟢		🟢		🟢
February		🟢		🟢		🟢		🟢		🟢
January		🟢		🟢		🟢		🟢		🟢
March		🟢		🟢		🟢		🟢		🟢
Fourth	188,170.24	🟢	98,853.98	🟢	192,174.31	🟢	1,319,738.96	🟢		🟢
December	96,931.78	🟢	22,948.65	🟢	55,327.10	🟢	478,092.84	🟢		🟢
2023-12-01 00:00:00.000		🟢		🟢		🟢	16,651.90	🟢		🟢
2023-12-02 00:00:00.000		🟢		🟢	5,624.74	🟢	22,950.52	🟢		🟢
2023-12-03 00:00:00.000		🟢		🟢		🟢	14,011.34	🟢		🟢
2023-12-04 00:00:00.000	5,906.30	🟢		🟢	8,737.71	🟢	22,224.66	🟢		🟢
2023-12-05 00:00:00.000	5,368.90	🟢	994.28	🟢	1,819.21	🟢	16,438.88	🟢		🟢
2023-12-06 00:00:00.000	7,113.18	🟢	1,118.56	🟢		🟢	1,467.55	🟢		🟢
2023-12-07 00:00:00.000		🟢		🟢	5,312.26	🟢	8,228.90	🟢		🟢
2023-12-08 00:00:00.000		🟢		🟢	2,033.23	🟢	7,001.47	🟢		🟢
2023-12-09 00:00:00.000	5,070.63	🟢		🟢	7,933.14	🟢	13,184.17	🟢		🟢
2023-12-10 00:00:00.000	9,276.72	🟢		🟢		🟢	12,417.26	🟢		🟢
2023-12-11 00:00:00.000	14,055.44	🟢	1,180.70	🟢		🟢	36,309.69	🟢		🟢
2023-12-12 00:00:00.000		🟢		🟢		🟢	11,114.26	🟢		🟢
2023-12-13 00:00:00.000		🟢	2,839.02	🟢		🟢	18,380.50	🟢		🟢
2023-12-14 00:00:00.000	6,025.28	🟢	1,658.32	🟢		🟢	13,321.23	🟢		🟢
2023-12-15 00:00:00.000		🟢		🟢		🟢	15,113.52	🟢		🟢
2023-12-16 00:00:00.000		🟢		🟢		🟢	12,594.63	🟢		🟢
2023-12-17 00:00:00.000	7,280.54	🟢		🟢		🟢	4,599.56	🟢		🟢
2023-12-18 00:00:00.000	2,800.82	🟢		🟢		🟢	9,324.50	🟢		🟢
2023-12-19 00:00:00.000	7,113.18	🟢		🟢	1,762.92	🟢	18,220.08	🟢		🟢
2023-12-20 00:00:00.000		🟢		🟢		🟢	13,796.50	🟢		🟢
2023-12-21 00:00:00.000	7,156.19	🟢		🟢		🟢	8,255.84	🟢		🟢
Total	1,130,804.67	🟢	906,205.71	🟢	1,085,240.88	🟢	11,179,634.94	🟢		🟢

Report 2 :

I implemented a three-tier cascading filter system based on the Geographical Hierarchy. By leveraging the relationships defined in the DimStore dimension of my SSAS cube, the slicers are context-aware. Selecting a Region automatically filters the City list, and selecting a City narrows the Store Name options. This design ensures a 'top-down' discovery path, allowing users to analyze performance at the provincial level before drilling into specific storefronts, preventing the selection of invalid data combinations.

1. First I added three slicers using the slicers using **Slicer Visual** . They are as below.
 - Region slicer – Graphical Hierarchy Region
 - Region.City slicer – Graphical Hierarchy City
 - Region.City.Store Name slicer – Graphical Hierarchy Store Name
2. Next I added a **Clear all slicer** button to clear the slicer if user want. This make this dashboard user friendly.



3. After that I added three **Card Visuals** to represent values of,
 - Total Amount
 - Total Cost
 - Total Sales (Fact Sales Count coulmn)

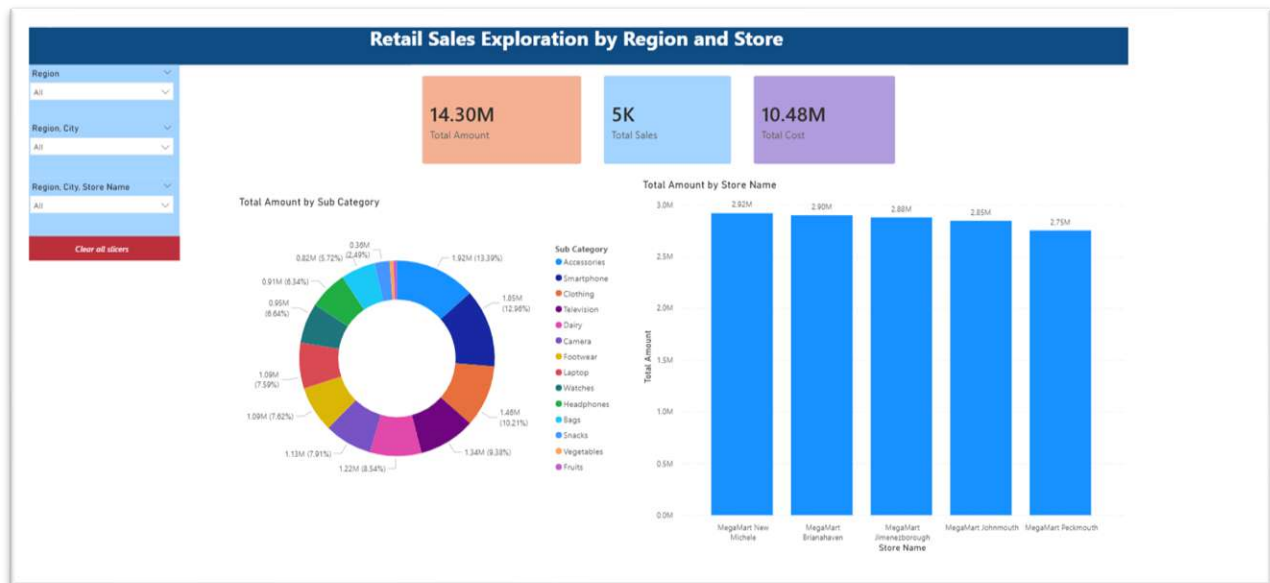


4. To represent **Total Amount** with **Sub Category** I used a **Donut Chart**. And added the following values to the fields.
 - Legend : Sub Category
 - Values : Total Amount

5. And also I added **Clustered Column Chart** to represent **Total Amount** with **Store Name** and added the following values to the fields.

- X – axis : Store Name
- Y – axis : Total Amount

6. Finally **Report 2** looked like below.



7. In there we can filter data using slicers like below.

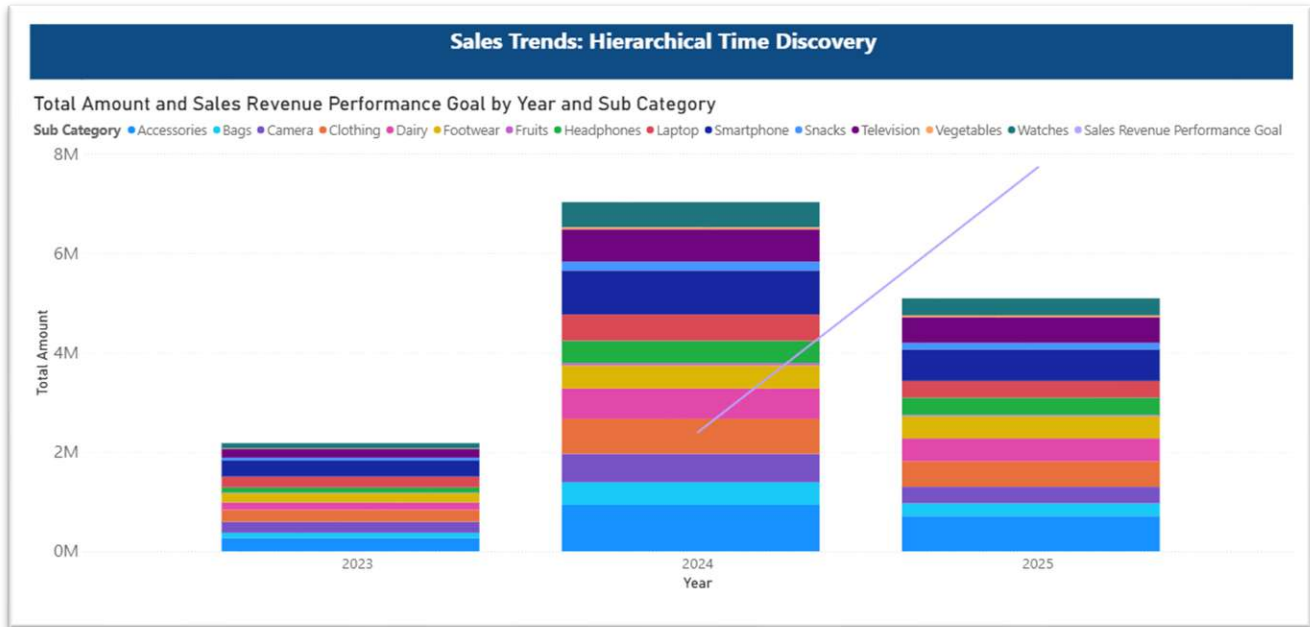
Region (East) > Region.City(Johnmouth)>Region.City.StoreName(Megamart Johnmouth)



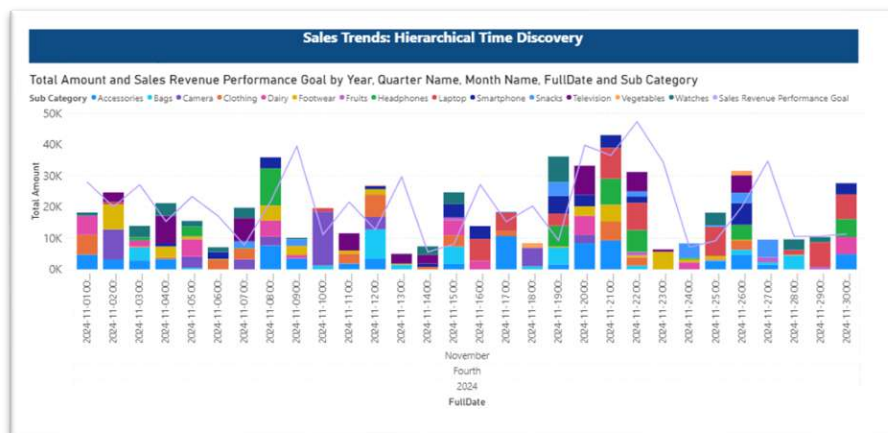
Report 3 :

I developed a drill-down report designed to explore the temporal dimension of the data warehouse, moving from macro-level annual summaries to micro-level monthly details. I utilized the **Calendar Hierarchy** as the primary X-axis. By enabling the 'Drill-Mode' feature, users can interactively navigate through the time levels without needing separate reports. The **Line and Stacked Column Chart** provides a dual-perspective, the columns show the composition of sales by 'Product Category,' while the line represents the sales trend. This allows for immediate identification of seasonal peaks or category-specific growth during the drill-down process.

1. I selected a **Line and Stacked Column Chart** from “Visualization” tab.
2. And next I configured the axis as below,
 - **X-Axis:** Dragged **Calendar Hierarchy** into the X-Axis bucket.
 - **Column Y-Axis:** Dragged Total Amount
 - **Line Y-Axis:** Dragged your **KPI Goal**
 - **Legend:** Dragged Product Category
3. Final look of the **Report 3** is as below.



4. By clicking on “Drill Down” button we can Drill down through the Calendar Hierarchy.



Report 4 :

I implemented a Drill-through report to facilitate deep-dive analysis, allowing users to navigate from a high-level executive summary to a granular, record-level detail page. I configured a hidden 'Detail' page with a **Drill-through Filter** mapped to the 'Product SubCategory' attribute of the SSAS cube. This enables context-aware navigation across report pages. By hiding the destination page, I ensured a clean report interface where detailed data is only presented when requested. The system automatically carries the filter context (SubCategory) from the source visual to the destination table, providing a seamless analytical experience.

- 1. First I created a new page as the target page and renamed it to “Product Detail view” and hide it.
- 2. Next I dragged **Product SubCategory** (from Dim Product) into the "Add drill-through fields here" box.
- 3. After I added following visuals to the target page.
 - **Table Visual:** Added Store Name, Product Name, Unit Price, and Total Amount.
 - **Card Visual:** Added Total Amount
- 4. Now the target table (Product Detail View) looked like this.



- 5. In a new page I created a **Treemap Visual** . And added the following fields.
 - **Category:** Product SubCategory.
 - **Values:** Total Amount



6. I did the Drill Through process as below.

- Go to the Summary Page.
- Right clicked on a specific slice of the Treemap (e.g., "Electronics").
- Hovered over **Drill-through** and select **Product Detail View**.
- Power BI jumped to the hidden page, and the entire page is already filtered to only show "Electronics."